

ICOHP Descriptors and Thermodynamic Stability Across a 394-Structure Dataset

Campaign 2: Dataset Construction, VASP+LOBSTER Computation, and Statistical Analysis

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Report generated as part of the `viability` project

August 16, 2026

The project’s central question is viability discrimination based on $\Delta(\text{ICOHP})$ (§6.4), across a 394-structure dataset.

Contents

1	Objective and Problem Statement	2
2	Methodology	2
2.1	Dataset Selection	2
2.2	Bonding-Type Classification	2
2.3	VASP + LOBSTER Computation	2
2.4	ICOHP/ICOB quantities: Integration, Cutoffs, Units	2
2.5	Statistics	3
3	Workflow Diagram	4
4	Environment	4
5	Dataset	4
6	Results	5
6.1	Correlations (Spearman) — classic ICOHP aggregates	5
6.2	Antibonding Population Near the Fermi Level	5
6.3	Reaction ICOHP	6
6.4	$\Delta(\text{ICOHP})$ -Based Viability Discrimination	7
7	Failures Found and Fixed	9
8	Compute Cost	9
9	Current Limits and Next Steps	10
10	Conclusion	10

A	Appendix: Percolation_viability	11
A.1	Original objective (mission #1)	11
A.2	Original workflow diagram	11
A.3	Campaign 2 dataset (186 compounds)	12
A.4	Results (with the percolation weight)	12
A.5	Mission #3: network dimensionality and periodic min-cut	13
A.6	Limits specific to the percolation weight	15
A.7	Conclusion of the original campaign	16
A.8	Compound List (percolation weight)	16
B	VASP Calculation Parameters	19
C	PAW-PBE Potentials	20
D	k-point Grid and NBANDS per Compound	21
E	Elements and Allotropes	25
F	Polymorphs and Allotropes	27
G	Target Compounds	30
H	Endobondic Reactions	30
I	Exobondic Reactions	31
J	ICOHP and ICOBI Antibonding Descriptors	33
J.1	Implementation	33
J.2	Correlation with formation energy	34
J.3	Δ (antibonding) of the decomposition-into-elements reaction	35

1 Objective and Problem Statement

Central research question (see §6.4): does $\Delta(\text{ICOHP})$ — the reaction-ICOHP descriptor for a compound’s decomposition into elements, and specifically the sign-based endobondic/exobondic classification it supports — distinguish thermodynamically stable, metastable, and unstable compounds?

2 Methodology

2.1 Dataset Selection

Materials Project query: unary and binary chemical systems, non-magnetic ($|\text{total magnetization}| \leq 0.01 \mu_B/\text{formula unit}$ — MP’s categorical `magnetic_ordering` field is "Unknown" for $\sim 98\%$ of entries and would have excluded nearly every genuinely non-magnetic compound), no f-block elements. The dataset combines two complementary selection tracks: a first set of 186 compounds (≤ 20 atoms/cell, 60 per stability family, a 2-entries-per-chemical-system cap to ensure elemental diversity rather than many polymorphs of the same system), and a second set of 163 compounds/elements (≤ 40 atoms/cell) specifically targeting additional elemental references, off-equilibrium polymorphs (high-pressure phases, allotropes), and alkali/alkaline-earth binaries against N/O/F/P/S/Cl, chosen to build genuine same-composition polymorph groups and to test descriptors far from equilibrium; and a third set of 45 binary compounds, each pairing an experimentally known, maximally metastable compound (farthest above the convex hull among its chemical system) with 1–3 theoretical polymorphs of the same formula sitting even farther above the hull, a robustness test deliberately targeting stability edge cases.

2.2 Bonding-Type Classification

Bond type (ionic / covalent / metallic) is computed by default by the `classify()` function (elemental composition + `is_metal`). A second, independent classification is built directly from each compound’s ICOHP/ICOB data (§5): `is_metal` for metallic character, then the mean ICOBI per bond (bond order, covalency) to separate ionic from covalent among non-metallic compounds. The two classifications are compared in §5.

2.3 VASP + LOBSTER Computation

Same template across all structures (static run, `ISYM=-1`, `LASPH=.TRUE.`, PAW-PBE potentials recommended by `pymatgen.io.vasp.sets.MPrelaxSet`): NBANDS is derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`), and basis-function determination goes through `Lobsterin.get_basis(potcar_symbols=...)` rather than direct POTCAR-file parsing (whose internal pymatgen validity check fails on several POTCARs unrecognized by hash in the local mirror).

2.4 ICOHP/ICOB Quantities: Integration, Cutoffs, Units

LOBSTER generation (`lobsterin`, identical across the dataset): pbeVaspFit2015 basis; COHP integration window from -35 to $+5$ eV around the Fermi level (`COHPstartEnergy/COHPendEnergy`); bond detection by interatomic distance from 0.1 to 6.0 Å (`cohpgenerator`, orbital-resolved); 0.05 eV Gaussian smearing (`gaussianSmearingWidth`).

Integrated ICOHP/ICOB (`ICOHPLIST.lobster/ICOBILIST.lobster`). A bond’s integrated ICOHP is, by LOBSTER’s own definition, the integral of $\text{COHP}(E)$ from the bottom of the window above up to the Fermi level (occupied states only), in eV; negative = bonding, positive = antibonding. ICOBI is the same integral over COBI (bond order index), dimensionless.

These are the per-bond, then per-compound-aggregated, quantities feeding §5 and the classic correlations of §6.

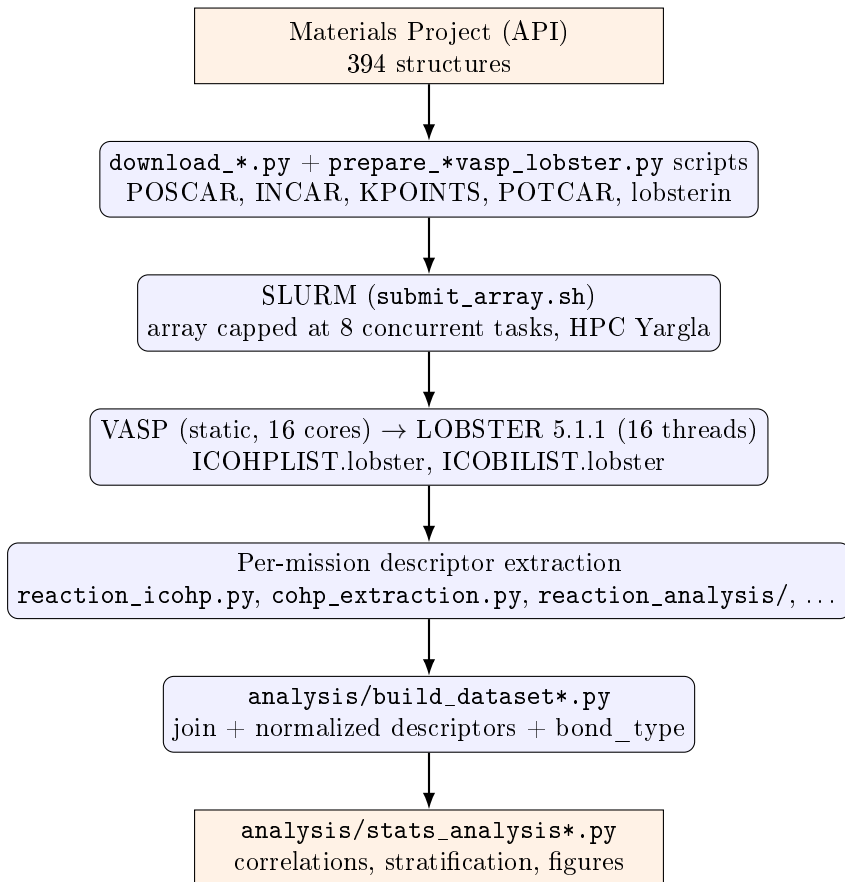
Reaction $\Delta(\text{ICOHP})$ (`reaction_analysis/`). For a balanced reaction (typically compound $\rightarrow$ elements), Δ = sum of products' ICOHP – sum of reactants' ICOHP, "products – reactants" convention, in three normalizations: per formula unit (extensive), per transferred atom (the normalization used throughout this report unless noted), and per bond (diagnostic only, non-conservative — bonds are not conserved between differently-bonded structures). Positive sign ("endobondic"): breaking the reactant's bonds costs more than the products' bonds recover — a possible kinetic barrier. Negative sign ("exobondic"): no such barrier is visible from bonding alone.

Antibonding population near the Fermi level (`cohpp_extraction.py`) — a quantity distinct from the $\Delta(\text{ICOHP})$ above. Rather than integrating the whole occupied COHP, this quantity integrates only its antibonding part ($\text{COHP} > 0$) over a one-sided window ($E_{\text{ref}} - \Delta E, E_{\text{ref}}$], where $E_{\text{ref}} = E_F$ (metallic compounds) or the VBM (gapped compounds), and $\Delta E \in \{0.5, 1.0, 2.0\}$ eV ($\Delta E = 1.0$ eV as the primary window) — only occupied states can destabilize the ground state via antibonding, hence the one-sided rather than symmetric window. Reported both raw (same "eV" unit convention as LOBSTER, though it is a Hamilton population integral, not literally an energy) and normalized by the total occupied ICOHP magnitude at the same E_{ref} . This quantity, and only this one, feeds the descriptor in §6.2 — the reaction $\Delta(\text{ICOHP})$ (§6.3–6.4) integrates the whole occupied COHP with no restriction to a window near E_F .

2.5 Statistics

Spearman correlation (no reason to expect a linear relationship) between each ICOHP/ICOBI descriptor and a stability target (`energy_above_hull` or `formation_energy_per_atom` depending on the descriptor), overall and per subgroup (`bond_type`, `is_metal`); any $n < 15$ subgroup is explicitly flagged, never presented bare. **No SISSO, no symbolic regression** at this stage (see §9).

3 Workflow Diagram



4 Environment

Identical across all 394 structures (Yargla, VASP 6.5.0, LOBSTER 5.1.1, Python 3.11 `.venv`), extended with: `mp-api`, `pandas`, `scipy`, `scikit-learn`, `matplotlib`, `pydantic`, `pyyaml` (see `requirements.txt`). Per-step core allocation: §8.

5 Dataset

The full set of computed compounds and elements totals **394 structures**.

Table 1: Dataset composition by element count.

Composition	<i>n</i>
Single-element references	68
Multi-element compounds	326
Total	394

Bond type. Two independent classifications (§2.2): the `classify()` function (elemental composition + `is_metal`) and a classification built directly from each compound’s ICOHP/ICOBI data (`is_metal` for metallic character, then a threshold on mean ICOBI per bond — calibrated at the midpoint of the median ICOBI of compounds `classify()` already labels ionic/covalent, i.e. 0.0370 — to separate ionic from covalent among non-metallic compounds).

Table 2: Stability class (by E_{hull} and experimental/theoretical character).

Class	n	E_{hull} range (eV/atom)
stable ($E_{\text{hull}} = 0$)	124	0.0000
experimental metastable	137	0.0000347 – 2.1080
theoretical metastable	131	0.0000843 – 3.1688
unknown (COD reference, no E_{hull})	2	—
Total	394	

Table 3: Bond type: `classify()` vs. the ICOHP/ICOB I classification (concordance on the subset `classify()` labels: 159/181 = 87.8%).

<code>classify()</code> \ ICOHP/ICOB I	metallic	ionic	covalent	unclassified
metallic	88	0	0	0
ionic	0	48	5	0
covalent	0	17	23	0
unclassified	138	33	39	1
Total	226	98	67	1

Of the 211 compounds `classify()` leaves "unclassified" (essentially alkali/alkaline-earth $\times$ {N, P, S} pairs outside `IONIC_ANIONS`, and metals bonded to an anion-like element that `classify()` deliberately excludes from its "metallic" bucket), the ICOHP/ICOB I classification resolves **72** into ionic (33) or covalent (39); the remaining 139 are either metallic (138, already covered by `is_metal`) or lack ICOB I data (1). Where both methods render a verdict, agreement is strong (87.8%) but not total: 22 ionic/covalent disagreements, worth checking case by case before using either classification as a stratification variable.

6 Results

6.1 Correlations (Spearman) — classic ICOHP aggregates

Table 4: Spearman correlations, classic ICOHP aggregates (sum, minimum) against E_{hull} , 186-compound subset.

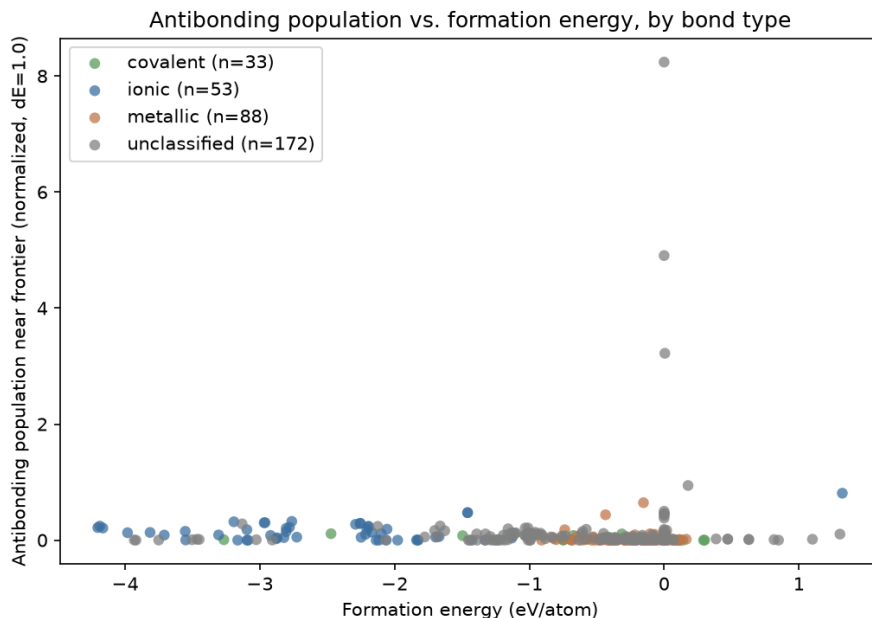
Group	Metric	n	ρ	p
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

These four rows serve as a reference baseline: raw ICOHP aggregates (per-compound sum, minimum) show little to no correlation with E_{hull} , except within the metallic subgroup. The descriptors built further on (antibonding population, reaction ICOHP, §6.2–§6.4) do markedly better, against `formation_energy_per_atom` rather than `energy_above_hull`.

6.2 Antibonding Population Near the Fermi Level

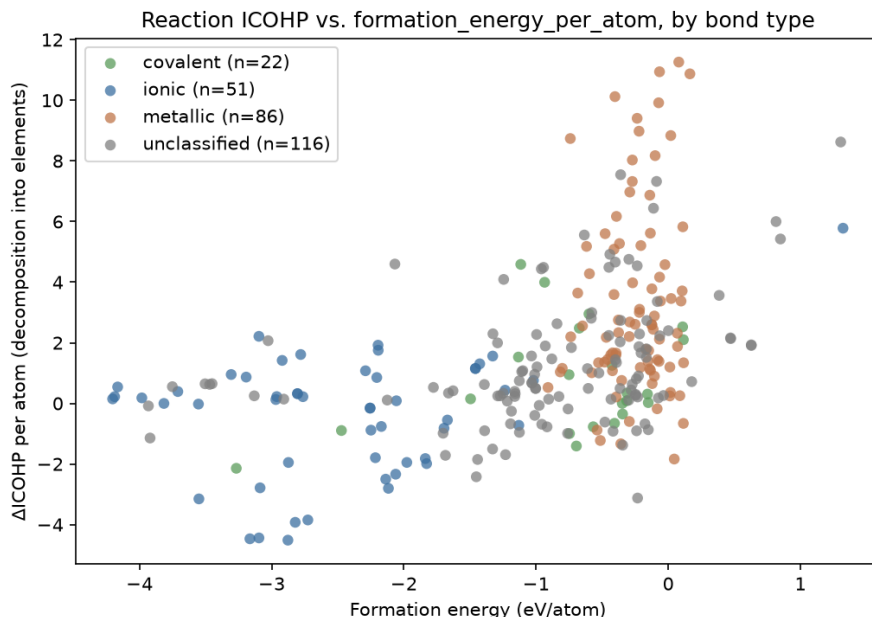
A question distinct from the fully-integrated ICOHP (§2.4): not *how much* bonding a compound has, but *how* its COHP is distributed in energy — the occupied antibonding fraction just

below the Fermi level/VBM, by analogy with Peierls/Jahn–Teller-type electronic instabilities. Result, against `formation_energy_per_atom`: $\rho = -0.282$, $p = 9.2 \times 10^{-8}$ ($n = 346$). The sign is the *opposite* of the naive Peierls/Jahn–Teller reading. The `bond_type=covalent` subgroup ($n = 33$) survives stratification clearly ($\rho = -0.551$, $p = 0.0009$) — the project’s most robust stratified result; `is_metal=False` also survives ($n = 157$, $p = 0.001$). An earlier result on the `bond_type=ionic` subgroup ($n = 9$, $\rho = -0.683$) does not replicate once the sample grows to $n = 53$ ($\rho = -0.184$, not significant) — a clear demonstration of the danger of $n < 15$ subgroups.



6.3 Reaction ICOHP

An ICOHP analog of `formation_energy_per_atom`: is a compound’s total ICOHP "cheaper" than that of the same atoms as reference elements? Result: $\rho = 0.502$, $p = 6.3 \times 10^{-19}$ ($n = 275$) — the dataset’s strongest and most robust continuous correlation, with an intuitive sign (less bonding than in elemental form $\Rightarrow$ less stable formation). `bond_type` stratification fully explains this layer of the pooled signal (Kruskal–Wallis $p = 1.1 \times 10^{-15}$, no stratum survives) — **but** `is_metal` stratification survives on both sides (`is_metal=False`: $\rho = 0.319$, $p = 0.0001$; `is_metal=True`: $\rho = 0.256$, $p = 0.0025$): between-group clustering fully explains the `bond_type`-level grouping, but not a real relationship within a fixed `is_metal`. Case 2 (direct same-formula polymorph comparison, 49 groups): 53.1% agreement between the most-bonded and the most-stable member, against a 45.9% chance baseline (exact Poisson-binomial test, $P = 0.19$) — no signal.



Still no SISSO. Reaction ICOHP is the dataset’s strongest raw correlate of `formation_energy_per_atom`, and the only descriptor whose signal is not fully explained away by between-group clustering at every level tested — the best argument yet for a stratified (`is_metal`-conditioned) rather than pooled feature search, still not attempted. Detailed reports: `analysis/REPORT_antibonding.md`, `analysis/REPORT_reaction_icohp.md`.

6.4 $\Delta(\text{ICOHP})$ -Based Viability Discrimination

Full detail: `analysis/REPORT_delta_icohp_viability.md`, `analysis/test_delta_icohp_viability.py`.

Reference validation. Per-bond-type ICOHP values for 7 reactions with independently known ΔICOHP (transcribed independently of any calculation in this project) are run end to end through `balance.py/delta.py/classify.py`:

Table 5: Validating the endobondic/exobondic classification on 7 reactions with independently known ΔICOHP .

Reaction	Reference (kJ/mol)	Computed (kJ/mol)	Label	
			Ref.	Computed
$\text{Pb}(\text{N}_3)_2 \rightarrow \text{Pb} + 3 \text{N}_2$	+1345	+1344.3	endo	endo
$\text{S}_4\text{N}_2 \rightarrow \frac{1}{2}\text{S}_8 + \text{N}_2$	+258	+258.5	endo	endo
$\text{S}_4\text{N}_4 \rightarrow \frac{1}{2}\text{S}_8 + 2\text{N}_2$	+399	+397.3	endo	endo
$\text{ZnSn} \rightarrow \text{Zn} + \text{Sn}$	−337	−336.6	exo	exo
$\text{CaO}[\text{sphalerite}] \rightarrow \text{CaO}[\text{rocksalt}]$	−79	−78.9	exo	exo
$\text{CaN} \rightarrow \frac{1}{3}\text{Ca}_3\text{N}_2 + \frac{1}{6}\text{N}_2$	−205	−204.8	exo	exo
$\text{Mn}_2\text{O}_7 \rightarrow 2\text{MnO}_2 + \frac{3}{2}\text{O}_2$	−186	−187.7	exo	exo

The 7 cases reproduce the reference ΔICOHP to within a few kJ/mol (source per-bond value rounding) and every `BondingLabel` matches. The Mn_2O_7 case is a useful guardrail on its own: exobondic under this static sign criterion, yet Mn_2O_7 is a real, observed compound that simply decomposes slowly (gradual O_2 loss) rather than through abrupt bond breaking — `classify.py` structurally attaches a warning to *every* `UNSTABLE_NONEXISTENT` verdict for exactly this reason, rather than special-casing Mn_2O_7 .

Correlation alone (Pearson/Spearman) is not the right test for "do our numbers match the reference values" — it is satisfied by any monotonic relationship, even with a large systematic offset or scale factor. **Lin’s concordance correlation coefficient** (CCC) specifically tests agreement with the identity line (computed = reference), the claim actually being tested here:

Table 6: Computed-vs-reference concordance on the 7 validation reactions.

Statistic	Value
n	7
Lin’s CCC	0.999998
Pearson r	0.999999 ($p = 3.5 \times 10^{-15}$)
Sign agreement	7/7
Mean residual	0.76 kJ/mol
Max residual	1.70 kJ/mol
RMSE	0.98 kJ/mol

A CCC this close to 1 means the two methods agree essentially to the reporting precision of the reference values themselves (rounded to 2–3 decimals), not merely that they move together — the implementation is validated, not just spot-checked.

Applied to the full dataset, pooled. 281 case-1 reactions, every element and compound with a computed decomposition-into-elements reaction, pooled into a single population (grouping by selection provenance would be exactly the kind of between-group confound this project’s methodology is meant to catch).

Table 7: Spearman correlations, Δ ICOHP/atom (case 1, 281 reactions) against two continuous stability targets.

Target	n	ρ	p
formation_energy_per_atom	275	−0.5016	6.3×10^{-19}
energy_above_hull_eV_per_atom	279	−0.1096	0.068

(Sign: `reaction_analysis`’s products–reactants convention, opposite to `reaction_icohp.py`’s own column §6.3 — same magnitude, flipped sign.) The formation-energy result is the same relationship as §6.3’s, confirmed here on the pooled population.

Sign-based discrimination (endobondic/exobondic). Experimental viability indicator (`theoretical=False` means experimentally realized):

Table 8: Endobondic/exobondic label against experimental/theoretical status.

	endobondic	exobondic
experimental ($n = 177$)	47	130
theoretical only ($n = 98$)	17	81

Fisher’s exact test: $p = 0.10$, odds ratio = 1.72 (endobondic is $\sim 1.7\times$ more frequent among experimentally realized compounds — the expected direction, endobondic being designed as a kinetic-barrier signature — but not significant at this sample size).

Kruskal–Wallis across the three groups: $H = 2.81$, $p = 0.245$ — **not significant**, but the ordering is exactly what physics predicts: experimentally metastable compounds (known to exist despite sitting off the convex hull — exactly the category the endobondic/exobondic framework addresses) have the highest endobondic fraction; never-synthesized theoretical metastable compounds have the lowest.

Table 9: Median $\Delta\text{ICOHP}/\text{atom}$ and % endobondic by stability class ($n = 175$).

Class	n	median $\Delta\text{ICOHP}/\text{atom}$ (eV)	% endobondic
experimental metastable	59	-1.077	27.1%
stable	59	-1.589	20.3%
theoretical metastable	59	-1.659	11.9%

Reading. The 7 reference reactions reproduce almost exactly — the implementation is correct. Extending the same sign-based logic to a much larger, more heterogeneous, and generally much more comfortably stable population does not yet produce a statistically significant stable/metastable/unstable discriminant, even though every trend points the right way. This is not a contradiction: the 7 reference reactions were specifically chosen as stability edge cases ($\text{Pb}(\text{N}_3)_2$, $\text{S}_4\text{N}_2/\text{S}_4\text{N}_4$, ZnSn , Mn_2O_7) where a kinetic-barrier explanation is specifically needed; most of this project’s 281 compounds are not — they are comfortably stable or comfortably not, where the thermodynamic driving force (the formation-energy correlation above) can simply dominate the sign-based kinetic signal. The full `ViabilityLabel` machinery in `classify.py` (beyond the sign-based `BondingLabel` alone used here) needs exactly such a near-stability-boundary population — for the current dataset, decomposition into elements is strongly exothermic for almost every compound (normal for stable inorganic chemistry), which would trivially classify nearly everything `STABLE_ON_HULL` and add no information beyond what `energy_above_hull` already says directly.

7 Failures Found and Fixed

Per the instruction ("every job failure must be logged with its probable cause"), 5 of the first 178 SLURM-array-submitted jobs failed on the first pass, all diagnosed and fixed at the source (the fixes, applied to the shared SLURM/INCAR templates, carry over to the whole dataset):

- **Al_{12}W , BW , WO_2 , TcW , TiW** (all tungsten-containing): the local mirror’s `W_sv` POT-CAR is missing its `LEXCH` field entirely (confirmed by direct inspection — present in every other variant checked, including plain `W`), which VASP reads as an incompatible exchange-correlation functional and refuses to run. Fixed by replacing the `W_sv` override with plain `W` (valid PBE, verified).
- **Al_{12}W** failed again after that fix, this time with a VASP SIGSEGV — a known issue (`ulimit -s unlimited` before `vasp_std`), present in `submit_array.sh` but forgotten in `prepare_vasp_lobster.py`’s per-compound SLURM template. Fixed; recovered on retry.

Final success rate across all 394 structures computed to date: **394/394**.

8 Compute Cost

394 structures computed in total: 68 single-element references and 326 multi-element compounds (§5).

Every VASP job, then LOBSTER job, runs on the **Yargla** HPC cluster, within a single SLURM allocation of one node (`-nodes=1`) and **16 cores**: VASP under MPI parallelism (`-ntasks=16 -cpus-per-task=1, srun`), then LOBSTER under OpenMP parallelism on the same 16 cores (`OMP_NUM_THREADS=16`, single process, run sequentially after VASP within the same job). The SLURM array (`submit_array.sh`) is capped at 8 concurrent tasks.

Combined sequential total (VASP + LOBSTER, 349 structures): **55.3 h**. Given the 8-concurrent-task cap, the order of magnitude for actual elapsed time to process this volume as a

Table 10: VASP and LOBSTER compute time (n=349; timing not available for every compound).

	VASP	LOBSTER
<i>n</i>	349	348*
Mean (s)	168	404
Median (s)	58	110
Max (s)	5715 (1.6 h)	12063 (3.4 h)
Sequential total	16.3 h	39.0 h

* 1 of 349 structures had no usable LOBSTER step (trivial-cell reference element).

single batch would be ~ 6.9 h — a rough estimate (dividing by 8, ignoring queue time) given for context.

9 Current Limits and Next Steps

- **classify() coverage:** 211/394 structures (54%) fall outside the ionic/covalent/metallic categories, mostly alkali/alkaline-earth $\times$ {N, P, S} bonds that `IONIC_ANIONS` does not recognize (§5) — purely informative here, but a real limitation for any mission that would want to stratify on it.
- **No true dynamical "unstable" class:** `theo_metastable` is a proxy (never synthesized), not a phonon-calculation result.
- **Per-subgroup sample size:** remains this project's structural limit at every new stratification (`bond_type`, `is_metal`) — see §6.2 for a concrete example where an $n < 15$ subgroup did not replicate once the sample grew.
- **Why not SISSO yet:** reaction ICOHP (§6.3) is now the strongest correlate and the first whose signal is not fully explained away by between-group clustering at every level tested — the best argument yet for a stratified feature search, still not attempted.

10 Conclusion

Reitz & Dronskowski's endobondic/exobondic methodology (§6.4) reproduces faithfully across the 7 independently known reference reactions (Lin's CCC 0.999998) and applies to this report's full dataset (281 decomposition-into-elements reactions). On that population, sign-based Δ ICOHP discrimination points the expected way (endobondic more frequent among experimental compounds, and among metastable rather than stable or never-synthesized compounds) but does not reach statistical significance at this scale (§6.4). Reaction ICOHP otherwise remains the dataset's strongest and most robust continuous correlate against formation energy ($\rho = 0.502$, §6.3).

Two concrete directions follow. First, deliberately extending to compounds near the stability boundary, where a kinetic-barrier explanation is specifically needed, rather than letting the dataset grow incidentally from whatever other analyses require — most current compounds are comfortably stable or comfortably unstable, a regime where the thermodynamic driving force simply dominates the sign-based kinetic signal. Second, a new descriptor: a reaction Δ (ICOHP) restricted to a window near the Fermi level, modeled on the antibonding population of §6.2 (§2.4) rather than on the fully integrated ICOHP — since it is precisely the near- E_F window that carries the project's most robust stratified subgroup, a "near- E_F " version of the reaction Δ ICOHP is a natural, not-yet-attempted extension.

Source code: <https://github.com/John-Akapulco/viability>

Data: `analysis/percolation_vs_formation_energy.csv`

Detailed reports: `analysis/REPORT_*.md`

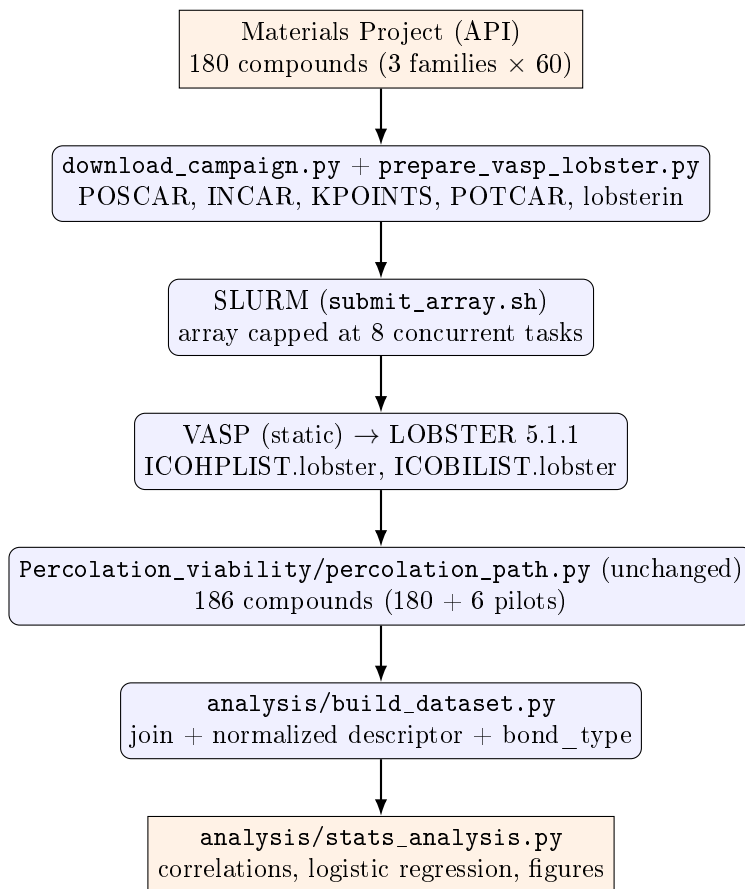
A Appendix: Percolation_viability

This appendix collects everything related to the percolation weight and its direct descendants (network dimensionality, periodic min-cut — missions #1 and #3): this was the central subject of the original campaign 2, unmodified below from its first drafting, but it is no longer this project's organizing question (§6.4). The corresponding source code has been moved to the repository's `Percolation_viability/` directory (`percolation_path.py`, `network_dimensionality.py`, `periodic_mincut.py`, and their tests `tests/test_percolation_path.py`, `tests/test_network_dimensionality.py`, `tests/test_periodic_mincut.py`).

A.1 Original objective (mission #1)

The very first report (6 toy compounds) established that the percolation descriptor computed by `percolation_path.py` diverges sharply from classic ICOHP aggregates (sum, mean, min, max), but without a sample large enough to statistically test whether that difference is *useful*: does the percolation descriptor carry predictive information about thermodynamic stability (`energy_above_hull`) that the classic aggregates don't already carry? Campaign 2 built a 186-compound dataset (180 new + the 6 pilot compounds) split into three families (`exp_stable`, `exp_metastable`, `theo_metastable`, §5) to settle that question.

A.2 Original workflow diagram



A.3 Campaign 2 dataset (186 compounds)

Family	n	E_{hull} range (eV/atom)
exp_stable	63	0.0000 (by construction)
exp_metastable	63	0.0005 – 0.0907
theo_metastable	60	0.0013 – 0.1991

Bond type (<code>classify()</code>)	n
metallic	88
covalent	23
ionic	9
unclassified	66

35% of compounds (66/186) fall outside `classify()`'s three categories — mostly transition-metal pnictides/chalcogenides and hydrogen-containing compounds. (Historical snapshot of campaign 2 alone; the up-to-date table across the current 349 structures is in §5.)

A.4 Results (with the percolation weight)

Group	Metric	n	ρ	p
all	percolation weight (raw)	186	0.064	0.383
all	percolation weight (normalized)	186	0.071	0.339
metallic	percolation weight (raw)	88	0.191	0.075
metallic	percolation weight (normalized)	88	0.222	0.038
ionic*	percolation weight (raw)	9	−0.402	0.284
covalent	percolation weight (raw)	23	0.165	0.453
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

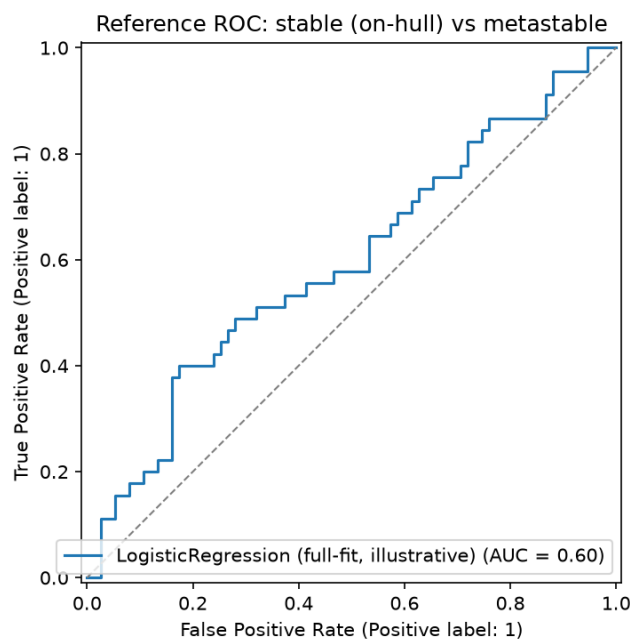
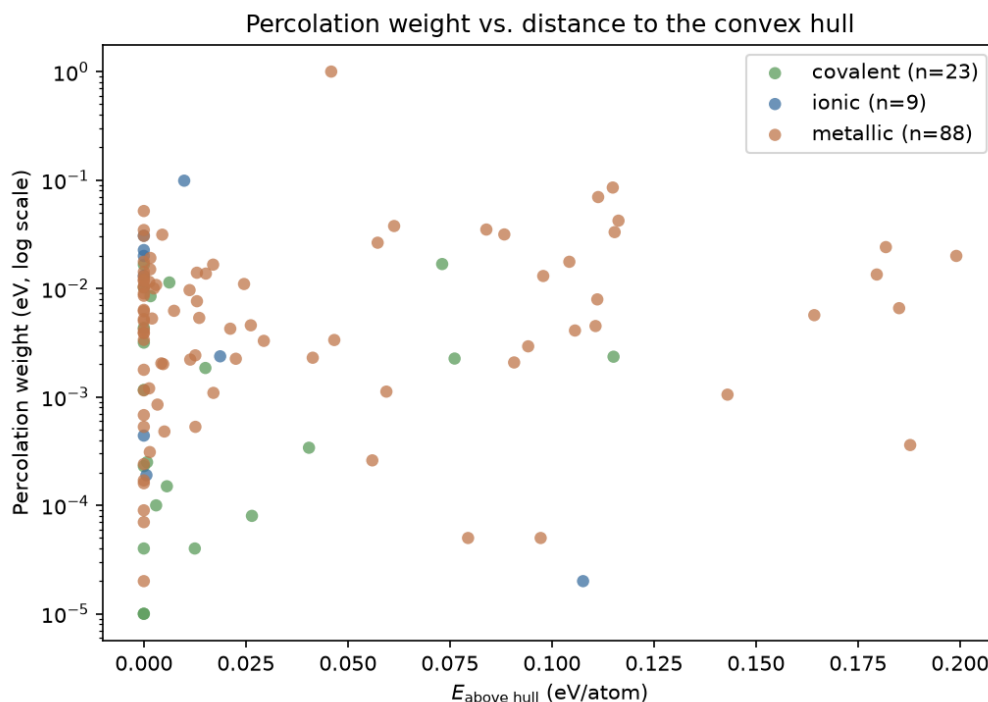
* $n < 15$: do not over-interpret.

In the one subgroup where any result reaches nominal significance (metallic, $n = 88$), the normalized percolation weight ($\rho = 0.222$, $p = 0.038$) and the ICOHP sum ($\rho = 0.223$, $p = 0.037$) are statistically indistinguishable in strength. **The percolation descriptor does not demonstrate additional predictive power over the classic aggregates on this dataset.**

Reference logistic regression. Stable (on-hull) vs. metastable, features: normalized percolation weight, mean ICOHP, bond-type dummies ($n = 120$ after dropping unclassified compounds).

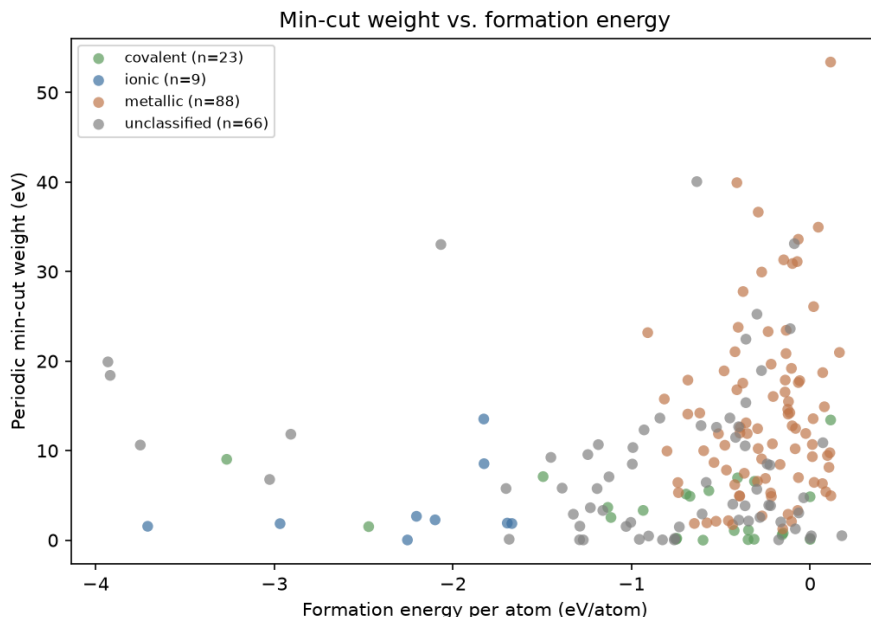
AUC (5-fold cross-validation)	0.496 ± 0.141
per-fold AUC	0.556, 0.430, 0.252, 0.637, 0.607

AUC ≈ 0.5 : chance level. The wide spread across folds (0.25–0.64) is itself informative: it indicates the absence of a stable, learnable signal at this sample size, not merely a "weak but real" one.



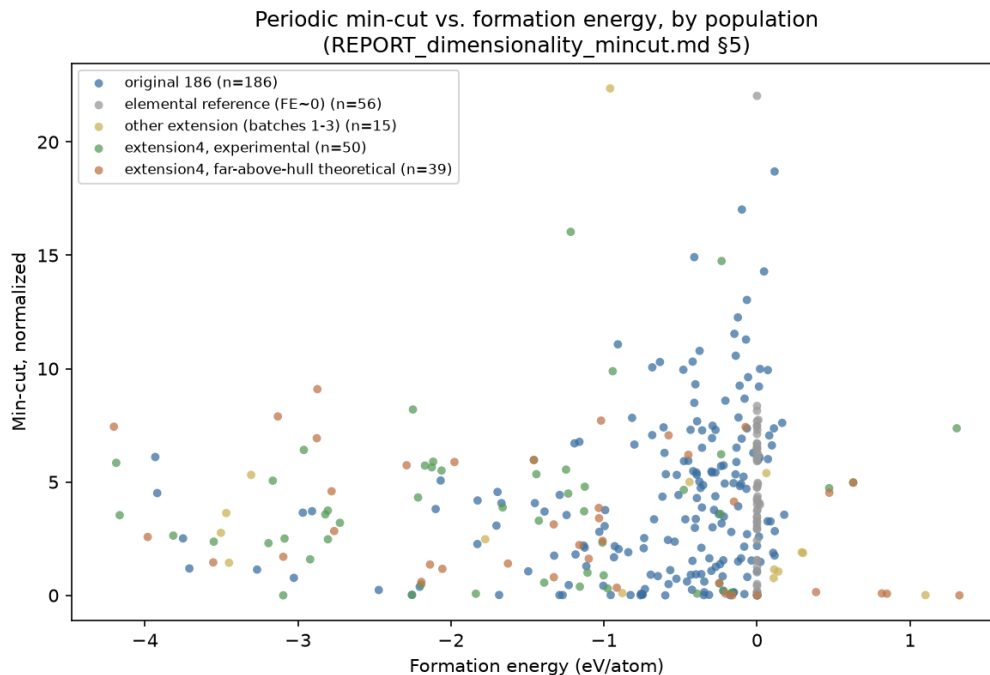
A.5 Mission #3: network dimensionality and periodic min-cut

Two new descriptors, `percolation_path.py` left unchanged: `network_dimensionality.py` (0D–3D classification via a relative bond-strength threshold + breadth-first search) and `periodic_mincut.py` (minimum cut via `networkx` max-flow, separability rather than traversability). Headline: the normalized min-cut is the **first descriptor in the project to reach a significant global correlation** ($\rho = 0.285$, $p = 1.0 \times 10^{-4}$, $n = 186$), likely driven in part by between-bond-type clustering; dimensionality alone does not separate `formation_energy_per_atom` (Kruskal–Wallis $p = 0.19$).



Why did the min-cut correlation collapse? Mission #3’s headline, $\rho = 0.285$ at $n = 186$, collapsed to $\rho = 0.089$, $p = 0.098$ (not significant) at $n = 349$. **This is not a refutation of the original result** — isolated, the original 186 compounds still give $\rho = 0.2845$, $p = 8.3 \times 10^{-5}$, essentially identical to the mission-#3 number. The collapse is a dilution artifact from pooling populations added for unrelated reasons:

- 65 of the 74 extension-batch-1–3 compounds are single-element reference structures (added for mission #5’s elemental references), sitting at `formation_energy_per_atom` ≈ 0 by construction with widely-scattered min-cut values — mechanically dilutes a rank correlation regardless of any true relationship. Excluding them recovers $\rho = 0.2157$ ($n = 193$).
- `extension4` is not one population but two, with *opposite-sign* correlations: the experimental half is flat ($\rho = 0.06$, not significant, $n = 50$); the deliberately far-above-hull theoretical half shows a *significant* correlation of the *opposite* sign ($\rho = -0.40$, $p = 0.011$, $n = 39$) — a higher min-cut there is associated with a *more* negative (more stable) formation energy, the reverse of the original population’s trend.



This opposite sign is itself resolved (August-16 follow-up, four independent checks on the 39 `theo_far_hull` compounds, `analysis/investigate_theo_far_hull_signflip.py`): neither a cell-size/symmetry selection artifact (survives controlling for `n_sites+sg_number`, $\rho = -0.39$) nor a real physical effect of far-from-hull structures, but a **between-anion-chemistry confound** — F-containing systems (high min-cut, strongly negative formation energy) and azide N_3 -containing systems (near-zero min-cut, near-zero/positive formation energy) pull the pooled correlation in opposite directions. Controlling for anion identity (dummy regression, $n = 39$) or comparing the experimental/theoretical pair within the exact same chemical system ($n = 37$ pairs) both independently collapse the correlation to $\rho \approx -0.07$, not significant — the same between-group clustering failure mode this project has now hit three times (min-cut/`bond_type`, antibonding/`is_metal`, and now min-cut/`anion`).

Recommendation: any future min-cut-vs-formation-energy number (or, by the same logic, any other descriptor's) should be conditioned on `family`/selection-provenance, the same way this project already conditions on `bond_type`/`is_metal`.

A.6 Limits specific to the percolation weight

- **Primitive vs. conventional cell:** not reprocessed on the 186-compound dataset. Most likely explanation for the weak observed correlations — the strongest bond can sit entirely inside the primitive cell and contribute to no non-contractile cycle at all, biasing the measured "weakest link." Tested on the 6 pilot compounds (mission #2): the choice changes the percolation weight substantially (3x-55x) but in the *opposite* direction from the hypothesis (the weight gets *smaller* with a larger cell); not extended to the full dataset based on this result.
- **Chance-level reference logistic regression** ($AUC \approx 0.5$) with percolation weight as a feature.
- **Why not SISSO:** a single candidate descriptor (the percolation weight, in two algebraically related variants), compared against aggregates that are themselves functions of the same underlying ICOHP values — no meaningful combinatorial search space at this stage.

A.7 Conclusion of the original campaign

The 186-compound dataset was built, computed (VASP+LOBSTER), and analyzed end to end. The headline result is a *negative* but quantified and honest one: at this sample size, the percolation descriptor shows no significant correlation with thermodynamic stability, and demonstrates no advantage over classic ICOHP aggregates. This result, together with the min-cut result (mission #3), motivated the project’s reframing around $\Delta(\text{ICOHP})$ documented in §6.4.

A.8 Compound List (percolation weight)

Campaign 2’s 186 compounds, grouped by category (family), with their Materials Project identifier, PBE thermodynamic hull distance (`energy_above_hull`), and the minimum ICOHP percolation weight (`icohp_percolation_weight_min`). Auto-generated from `analysis/percolation_vs_hull.csv` by `report/gen_appendix.py`.

Table S1: Complete list of the 186 compounds, grouped by category, with their Materials Project identifier, PBE hull distance, and minimum ICOHP percolation weight.

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
Experimental, stable (hull) ($n = 63$)				
As ₂ S ₃	mp-1189572	covalent	0.0000	1.00e-05
AsF ₃	mp-28027	covalent	0.0000	0.00e+00
BI ₃	mp-23189	covalent	0.0000	1.00e-05
BW	mp-7832	—	0.0000	1.90e-04
BaAu ₂	mp-30363	metallic	0.0000	5.19e-02
BaCl ₂	mp-568662	ionic	0.0000	1.31e-02
BaH ₂	mp-23715	—	0.0000	2.40e-04
Be ₂ Cu	mp-2031	metallic	0.0000	1.15e-03
Be ₂ Nb ₃	mp-11272	metallic	0.0000	6.36e-03
Be ₂ V	mp-11281	metallic	0.0000	6.80e-04
CaAl ₄	mp-1749	metallic	0.0000	1.03e-02
CaSn	mp-2450	metallic	0.0000	5.23e-03
CdPd	mp-1696	metallic	0.0000	9.00e-05
CoB	mp-20857	—	0.0000	6.00e-05
Cs ₂ As ₃	mp-15557	—	0.0000	2.02e-02
FeS	mp-505531	—	0.0000	3.05e-03
Ge ₂ Os	mp-10032	metallic	0.0000	1.21e-02
HfTc ₂	mp-1095669	metallic	0.0000	1.77e-02
Hg ₄ Pt	mp-936	metallic	0.0000	3.36e-03
HgF ₂	mp-8177	—	0.0000	8.95e-03
InBr	mp-22870	covalent	0.0000	1.02e-02
KI	mp-22898	ionic	0.0000	2.26e-02
KN ₃	mp-827	—	0.0000	2.80e-04
Li ₂ Se	mp-2286	—	0.0000	1.16e-02
LiB	mp-1001835	—	0.0000	2.91e-03
Mg ₂ Ni	mp-2137	metallic	0.0000	1.60e-04
MgP ₄	mp-384	—	0.0000	0.00e+00
MnAl ₁₂	mp-1104165	metallic	0.0000	1.42e-02
MoSe ₂	mp-1634	covalent	0.0000	1.16e-03
NaS	mp-2400	—	0.0000	0.00e+00
Na	mp-10172	—	0.0000	1.97e-03
Nb ₃ Ir	mp-1458	metallic	0.0000	9.02e-03
NbGa ₃	mp-1973	metallic	0.0000	8.58e-03
PI ₂	mp-29443	covalent	0.0000	2.30e-04
PbSe	mp-2201	covalent	0.0000	1.66e-02

Table S1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
RbI	mp-22903	ionic	0.0000	2.00e-02
Sc5Ge3	mp-17190	metallic	0.0000	5.30e-04
ScAg2	mp-1018128	metallic	0.0000	1.70e-04
ScTe	mp-10026	—	0.0000	7.12e-03
SnPt	mp-19856	metallic	0.0000	1.29e-02
SrO2	mp-2697	ionic	0.0000	4.40e-04
SrSi	mp-2661	metallic	0.0000	7.00e-05
Ta5Ge3	mp-17593	metallic	0.0000	3.46e-02
TaP	mp-1067587	—	0.0000	3.14e-02
TaSb2	mp-11697	metallic	0.0000	1.17e-02
Te2Pd	mp-782	—	0.0000	1.44e-02
Te2Ru	mp-267	covalent	0.0000	4.34e-03
TeO2	mp-8377	covalent	0.0000	4.00e-05
Ti3In	mp-866046	metallic	0.0000	5.01e-03
Ti3Pt	mp-2339	metallic	0.0000	1.78e-03
TiBe12	mp-1104067	metallic	0.0000	2.40e-04
TiNi	mp-1048	metallic	0.0000	3.91e-03
TiO	mp-1071163	—	0.0000	6.40e-04
TiSi2	mp-1077503	metallic	0.0000	6.18e-03
TlCl	mp-571079	—	0.0000	7.17e-03
VNi2	mp-11531	metallic	0.0000	3.88e-03
WO2	mp-1524394	—	0.0000	2.20e-04
Zr2Au	mp-2348	metallic	0.0000	3.06e-02
Zr5Pb3	mp-681992	metallic	0.0000	2.00e-05
ZrMo2	mp-2049	metallic	0.0000	1.05e-02
Si	mp-149	covalent	0.0000	3.17e-03
NaCl	mp-22862	ionic	0.0000	3.07e-02
AlNi	mp-1487	metallic	0.0000	4.19e-03
Experimental, metastable ($n = 63$)				
CdI2	mp-567259	—	0.0005	2.38e-02
MgCl2	mp-570259	ionic	0.0006	1.90e-04
C	mp-169	covalent	0.0008	2.50e-04
ZrBr	mp-1065846	—	0.0008	7.88e-03
Li3Hg	mp-1646	metallic	0.0013	1.20e-03
Ta7Co6	mp-1104548	metallic	0.0015	3.10e-04
CaGa4	mp-2461	metallic	0.0016	1.91e-02
InCl	mp-571636	covalent	0.0016	8.50e-03
NiB	mp-14019	—	0.0018	4.10e-04
MgH2	mp-23711	—	0.0019	1.00e-05
Sn7Ir5	mp-20466	metallic	0.0024	1.00e-02
NaN3	mp-1066400	—	0.0029	7.00e-04
ClF3	mp-556767	—	0.0030	0.00e+00
SiS2	mp-1095294	covalent	0.0030	1.00e-04
Sr5Sn3	mp-17720	metallic	0.0030	1.08e-02
Te2Ir	mp-1551	—	0.0030	2.80e-03
Li13Sn5	mp-30769	metallic	0.0033	8.50e-04
H3N	mp-643432	covalent	0.0038	0.00e+00
Cs2Se	mp-1011696	—	0.0042	2.62e-02
In3Ir	mp-630976	metallic	0.0043	2.04e-03
HI	mp-697025	—	0.0044	0.00e+00
Rb2Hg7	mp-31474	metallic	0.0045	3.14e-02
PbAu2	mp-19871	metallic	0.0047	2.01e-03
BeCu	mp-2323	metallic	0.0050	4.80e-04
SiO2	mp-546794	covalent	0.0056	1.50e-04

Table S1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
IF7	mp-27988	—	0.0063	0.00e+00
K	mp-10157	—	0.0069	8.07e-02
Al3Os2	mp-16521	metallic	0.0074	6.23e-03
Nb2O5	mp-1595	—	0.0075	1.00e-04
LiBr	mp-23259	ionic	0.0099	9.88e-02
ZnAg	mp-1912	metallic	0.0112	9.67e-03
MgPd3	mp-7753	metallic	0.0114	2.21e-03
AsPd2	mp-1080831	—	0.0118	3.50e-04
GeSe2	mp-10074	covalent	0.0125	4.00e-05
Re3Ge7	mp-29141	metallic	0.0126	2.42e-03
Cu3Ge	mp-19724	metallic	0.0126	5.30e-04
Ti3Ge	mp-1189044	metallic	0.0130	1.40e-02
ZrGa3	mp-30685	metallic	0.0130	7.64e-03
SrAl2	mp-1638	metallic	0.0136	5.36e-03
NiS	mp-1547	—	0.0142	3.00e-04
Al12W	mp-11227	metallic	0.0152	1.37e-02
CsH	mp-632319	—	0.0161	2.66e-02
CaGa2	mp-992	metallic	0.0170	1.66e-02
NiHg4	mp-570835	metallic	0.0171	1.09e-03
SiRh	mp-2287	metallic	0.0246	1.10e-02
BaSn5	mp-571353	metallic	0.0262	4.58e-03
PS	mp-612	covalent	0.0265	8.00e-05
AsRh	mp-22079	—	0.0269	1.93e-03
Te2Au	mp-567525	—	0.0274	1.62e-02
Be3N2	mp-6977	—	0.0315	2.00e-05
Y2O3	mp-558573	—	0.0342	5.00e-05
AgO	mp-1079720	—	0.0423	4.00e-05
FeTe2	mp-1102002	—	0.0449	5.20e-04
H2O	mp-2739191	—	0.0494	0.00e+00
CdO2	mp-2310	—	0.0562	9.00e-05
Mo3Se4	mp-21021	—	0.0609	1.84e-03
PbSe	mp-22009	covalent	0.0731	1.68e-02
Be12Pd	mp-12666	metallic	0.0795	5.00e-05
Sr2As	mp-15697	—	0.0800	2.06e-03
AsRh2	mp-1102364	—	0.0806	4.63e-03
ZrRh	mp-2808	metallic	0.0839	3.50e-02
K2S	mp-559278	—	0.0873	1.07e-02
Ge4Rh	mp-1104286	metallic	0.0907	2.08e-03
Theoretical, metastable ($n = 60$)				
Li3Ag	mp-976408	metallic	0.0013	1.15e-02
Y3Co	mp-1189329	metallic	0.0016	1.50e-02
TiNi	mp-1067248	metallic	0.0020	5.29e-03
TaP	mp-1187244	—	0.0033	3.59e-02
Sr3As4	mp-678007	—	0.0062	4.47e-03
Ga2Se3	mp-1224809	covalent	0.0063	1.14e-02
MnN	mp-1009130	covalent	0.0151	1.85e-03
BeCl2	mp-1190080	ionic	0.0187	2.37e-03
MgSn	mp-1094801	metallic	0.0212	4.26e-03
TiW	mp-1216621	metallic	0.0225	2.25e-03
SrTi3	mp-1206636	metallic	0.0294	3.29e-03
IrCl3	mp-729507	—	0.0301	2.72e-03
Ta4C3	mp-1218000	—	0.0335	1.44e-03
Sc2O3	mp-755313	—	0.0364	5.10e-04
SnBr2	mp-978985	covalent	0.0405	3.40e-04

Table S1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
Sn7Pd5	mp-1219006	metallic	0.0414	2.30e-03
Tl4Sn	mp-1216560	metallic	0.0459	1.00e+00
LiTi3	mp-1185253	metallic	0.0467	3.35e-03
PtCl2	mp-1219748	—	0.0490	5.04e-03
HCl	mp-684609	—	0.0536	2.50e-04
MgSn	mp-1094669	metallic	0.0560	2.60e-04
K5As4	mp-684799	—	0.0564	8.80e-03
KSn3	mp-1185151	metallic	0.0573	2.65e-02
NbS2	mp-774873	—	0.0576	1.20e-04
YAg3	mp-1187811	metallic	0.0594	1.12e-03
B3Ir2	mp-1228647	—	0.0609	2.80e-04
TiMo2	mp-1216675	metallic	0.0613	3.77e-02
Cr2C	mp-1226378	—	0.0625	2.08e-03
ZrO2	mp-775909	—	0.0627	1.20e-04
Zn	mp-2647117	—	0.0717	1.04e-02
IN2	mp-1097011	—	0.0730	4.70e-04
CoSe2	mp-1390196	—	0.0734	8.40e-04
GeS	mp-12910	covalent	0.0762	2.26e-03
ZrTi	mp-1215236	metallic	0.0883	3.15e-02
VCo3	mp-1095057	metallic	0.0942	2.93e-03
Tl3Ga	mp-1187539	metallic	0.0972	5.00e-05
CaGe2	mp-643542	metallic	0.0979	1.30e-02
NbRh2	mp-1220414	metallic	0.1043	1.76e-02
TcW	mp-1217337	metallic	0.1056	4.10e-03
MgF2	mp-560236	ionic	0.1076	2.00e-05
VN	mp-1017532	—	0.1080	4.00e-05
Na2Cl	mp-990084	—	0.1105	3.76e-02
ZnPb3	mp-1187949	metallic	0.1107	4.51e-03
Sr3Sn	mp-1187136	metallic	0.1111	7.94e-03
TaRe	mp-1217894	metallic	0.1113	6.98e-02
Al4Si	mp-1228120	metallic	0.1149	8.54e-02
B6Pb	mp-1096825	covalent	0.1151	2.35e-03
ScTc3	mp-867262	metallic	0.1154	3.31e-02
CrMo	mp-1226200	metallic	0.1163	4.23e-02
BPd5	mp-1228665	—	0.1325	2.42e-02
Zr3Be	mp-1188016	metallic	0.1430	1.05e-03
Na3Cl	mp-1064484	—	0.1456	2.55e-03
SiAu3	mp-1186998	metallic	0.1643	5.68e-03
InSe2	mp-1232036	—	0.1733	1.64e-03
TiSi2	mp-570991	metallic	0.1796	1.35e-02
BeAu3	mp-983539	metallic	0.1818	2.41e-02
Zr3Ge	mp-1188039	metallic	0.1851	6.59e-03
AlSb	mp-1023918	metallic	0.1878	3.60e-04
Mg15B	mp-1023515	—	0.1943	2.80e-04
HfMo	mp-1224314	metallic	0.1991	2.00e-02

B VASP Calculation Parameters

Fixed INCAR parameters, identical across the whole dataset (static, LOBSTER-compatible run) — only NBANDS (appendix D) and the k-point grid vary by compound.

Parameter	Value	Description
PREC	Accurate	Overall precision (charge grid, projectors)
ENCUT	600.0	Plane-wave cutoff energy (eV)
EDIFF	1e-06	Electronic convergence criterion (eV)
IBRION	-1	Ionic dynamics type (-1 = static, no relaxation)
NSW	0	Number of ionic steps (0 = static run on the MP structure)
ISMear	0	Smearing type (0 = Gaussian)
SIGMA	0.05	Smearing width (eV)
LASPH	True	Non-spherical corrections to the PAW potential
ISYM	-1	Symmetry (-1 = disabled, required by LOBSTER)
LWAVE	True	Write WAVECAR (required by LOBSTER)
LCHARG	True	Write CHGCAR
NPAR	4	Band-level parallelization
KPAR	2	K-point-level parallelization

C PAW-PBE Potentials

One PAW-PBE potential per element present in the dataset (pymatgen’s `MPRelaxSet` library — the same one Materials Project used to relax the input structures), with one documented exception for tungsten (§7): the recommended `W_pv` variant is absent from the local mirror, and `W_sv` turned out to be corrupted (missing `LEXCH` field) — plain `W` is used instead, verified valid.

Table S2: PAW-PBE potentials used, one per element present in the dataset (pymatgen’s `MPRelaxSet` library, with one documented exception: `W` substituted for `W_pv`/`W_sv`, see §7).

Element	PAW-PBE potential
Ag	Ag (02Apr2005)
Al	Al (04Jan2001)
As	As (22Sep2009)
Au	Au (04Oct2007)
B	B (06Sep2000)
Ba	Ba_sv (06Sep2000)
Be	Be_sv (06Sep2000)
Br	Br (06Sep2000)
C	C (08Apr2002)
Ca	Ca_sv (06Sep2000)
Cd	Cd (06Sep2000)
Cl	Cl (06Sep2000)
Co	Co (02Aug2007)
Cr	Cr_pv (02Aug2007)
Cs	Cs_sv (25Jan2019)
Cu	Cu_pv (06Sep2000)
F	F (08Apr2002)
Fe	Fe_pv (02Aug2007)
Ga	Ga_d (06Jul2010)
Ge	Ge_d (03Jul2007)
H	H (15Jun2001)
Hf	Hf_pv (06Sep2000)

Element	PAW-PBE potential
Hg	Hg (06Sep2000)
I	I (08Apr2002)
In	In_d (06Sep2000)
Ir	Ir (06Sep2000)
K	K_sv (06Sep2000)
Li	Li_sv (10Sep2004)
Mg	Mg_pv (13Apr2007)
Mn	Mn_pv (02Aug2007)
Mo	Mo_pv (04Feb2005)
N	N (08Apr2002)
Na	Na_pv (19Sep2006)
Nb	Nb_pv (08Apr2002)
Ni	Ni_pv (06Sep2000)
O	O (08Apr2002)
Os	Os_pv (20Jan2003)
P	P (06Sep2000)
Pb	Pb_d (06Sep2000)
Pd	Pd (04Jan2005)
Pt	Pt (04Feb2005)
Rb	Rb_sv (06Sep2000)
Re	Re_pv (06Sep2000)
Rh	Rh_pv (25Jan2005)
Ru	Ru_pv (28Jan2005)
S	S (06Sep2000)
Sb	Sb (06Sep2000)
Sc	Sc_sv (07Sep2000)
Se	Se (06Sep2000)
Si	Si (05Jan2001)
Sn	Sn_d (06Sep2000)
Sr	Sr_sv (07Sep2000)
Ta	Ta_pv (07Sep2000)
Tc	Tc_pv (04Feb2005)
Te	Te (08Apr2002)
Ti	Ti_pv (07Sep2000)
Tl	Tl_d (06Sep2000)
V	V_pv (07Sep2000)
W	W (08Apr2002)
Y	Y_sv (25May2007)
Zn	Zn (06Sep2000)
Zr	Zr_sv (04Jan2005)

D k-point Grid and NBANDS per Compound

k-point grid auto-generated by `pymatgen.io.vasp.Kpoints.automatic_density_by_vol` (density `kppv1=100`, Γ -centered or Monkhorst–Pack mesh depending on cell symmetry) and `NBANDS`, derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`) — a strictly sufficient value for the local-basis projection, not a fixed guess.

Table S3: k-point grid (pymatgen automatic density, `kppvol=100`) and number of bands (derived from the LOBSTER basis) per compound.

Formula	mp-id	k-point grid	NBANDS
Experimental, stable (hull) ($n = 63$)			
As ₂ S ₃	mp-1189572	5 3 2 (G)	80
AsF ₃	mp-28027	5 4 4 (G)	64
BI ₃	mp-23189	4 4 4 (G)	32
BW	mp-7832	9 9 3 (G)	52
BaAu ₂	mp-30363	6 6 7 (G)	23
BaCl ₂	mp-568662	6 6 6 (G)	13
BaH ₂	mp-23715	6 4 3 (G)	28
Be ₂ Cu	mp-2031	7 7 7 (G)	38
Be ₂ Nb ₃	mp-11272	8 4 4 (M)	74
Be ₂ V	mp-11281	7 7 4 (G)	76
CaAl ₄	mp-1749	7 7 4 (G)	21
CaSn	mp-2450	6 6 4 (M)	28
CdPd	mp-1696	7 6 6 (G)	36
CoB	mp-20857	9 7 5 (G)	52
Cs ₂ As ₃	mp-15557	4 4 3 (G)	44
FeS	mp-505531	8 8 5 (G)	26
Ge ₂ Os	mp-10032	9 6 4 (G)	54
HfTc ₂	mp-1095669	5 5 3 (G)	108
Hg ₄ Pt	mp-936	5 5 5 (G)	45
HgF ₂	mp-8177	8 8 8 (G)	17
InBr	mp-22870	6 6 4 (M)	26
KI	mp-22898	6 6 6 (G)	9
KN ₃	mp-827	5 5 5 (G)	34
Li ₂ Se	mp-2286	7 7 7 (G)	14
LiB	mp-1001835	7 7 9 (G)	18
Mg ₂ Ni	mp-2137	5 5 2 (G)	102
MgP ₄	mp-384	5 5 3 (G)	40
MnAl ₁₂	mp-1104165	4 4 4 (M)	57
MoSe ₂	mp-1634	9 9 2 (G)	34
NaS	mp-2400	6 6 3 (G)	32
Na	mp-10172	8 8 4 (G)	8
Nb ₃ Ir	mp-1458	5 5 5 (G)	72
NbGa ₃	mp-1973	8 8 6 (M)	36
PI ₂	mp-29443	6 4 3 (G)	24
PbSe	mp-2201	7 7 7 (G)	13
RbI	mp-22903	6 6 6 (G)	9
Sc ₅ Ge ₃	mp-17190	3 3 5 (G)	154
ScAg ₂	mp-1018128	8 8 5 (G)	28
ScTe	mp-10026	7 7 4 (G)	28
SnPt	mp-19856	7 7 5 (G)	36
SrO ₂	mp-2697	8 8 7 (G)	13
SrSi	mp-2661	7 6 4 (G)	18
Ta ₅ Ge ₃	mp-17593	4 4 4 (M)	144
TaP	mp-1067587	9 9 4 (G)	26
TaSb ₂	mp-11697	8 5 4 (G)	34
Te ₂ Pd	mp-782	7 7 5 (G)	17
Te ₂ Ru	mp-267	7 5 4 (G)	34
TeO ₂	mp-8377	6 5 3 (G)	48
Ti ₃ In	mp-866046	5 5 6 (G)	72
Ti ₃ Pt	mp-2339	5 5 5 (G)	72
TiBe ₁₂	mp-1104067	7 5 5 (G)	69

Table S3 (continued)

Formula	mp-id	k-point grid	NBANDS
TiNi	mp-1048	10 7 6 (G)	36
TiO	mp-1071163	10 6 6 (G)	39
TiSi2	mp-1077503	8 8 4 (M)	34
TlCl	mp-571079	6 6 4 (M)	26
VNi2	mp-11531	12 8 7 (G)	27
WO2	mp-1524394	6 5 5 (G)	68
Zr2Au	mp-2348	9 9 4 (G)	29
Zr5Pb3	mp-681992	3 3 5 (G)	154
ZrMo2	mp-2049	6 6 6 (G)	56
Si	mp-149	8 8 8 (G)	100
NaCl	mp-22862	8 8 8 (G)	100
AlNi	mp-1487	10 10 10 (M)	100
Experimental, metastable ($n = 63$)			
CdI2	mp-567259	7 7 4 (G)	17
MgCl2	mp-570259	8 8 5 (G)	12
C	mp-169	12 12 8 (M)	100
ZrBr	mp-1065846	8 8 3 (G)	28
Li3Hg	mp-1646	7 7 7 (G)	24
Ta7Co6	mp-1104548	6 6 3 (G)	117
CaGa4	mp-2461	7 7 5 (G)	41
InCl	mp-571636	6 6 4 (M)	26
NiB	mp-14019	10 10 7 (G)	26
MgH2	mp-23711	6 5 5 (G)	24
Sn7Ir5	mp-20466	4 4 4 (M)	108
NaN3	mp-1066400	7 7 7 (G)	16
ClF3	mp-556767	6 4 3 (G)	64
SiS2	mp-1095294	5 4 3 (G)	48
Sr5Sn3	mp-17720	3 3 3 (G)	104
Te2Ir	mp-1551	7 5 2 (G)	51
Li13Sn5	mp-30769	6 6 1 (G)	110
H3N	mp-643432	8 5 5 (G)	28
Cs2Se	mp-1011696	5 3 2 (G)	56
In3Ir	mp-630976	4 4 4 (M)	144
HI	mp-697025	3 3 3 (G)	40
Rb2Hg7	mp-31474	4 4 4 (G)	73
PbAu2	mp-19871	5 5 5 (G)	54
BeCu	mp-2323	10 10 10 (M)	100
SiO2	mp-546794	6 6 6 (M)	24
IF7	mp-27988	5 5 3 (G)	64
K	mp-10157	6 6 6 (G)	5
Al3Os2	mp-16521	9 9 3 (G)	30
Nb2O5	mp-1595	7 7 4 (G)	38
LiBr	mp-23259	8 8 8 (G)	100
ZnAg	mp-1912	9 9 9 (G)	18
MgPd3	mp-7753	7 7 7 (G)	31
AsPd2	mp-1080831	8 4 4 (G)	66
GeSe2	mp-10074	5 5 5 (G)	34
Re3Ge7	mp-29141	9 6 1 (G)	180
Cu3Ge	mp-19724	6 6 5 (G)	72
Ti3Ge	mp-1189044	4 4 4 (M)	144
ZrGa3	mp-30685	7 7 5 (G)	37
SrAl2	mp-1638	5 5 5 (G)	26
NiS	mp-1547	6 6 6 (M)	39
Al12W	mp-11227	4 4 4 (M)	57

Table S3 (continued)

Formula	mp-id	k-point grid	NBANDS
CsH	mp-632319	7 7 7 (G)	6
CaGa2	mp-992	7 7 6 (G)	23
NiHg4	mp-570835	6 6 6 (M)	45
SiRh	mp-2287	6 6 5 (G)	52
BaSn5	mp-571353	5 5 4 (G)	50
PS	mp-612	4 4 3 (G)	64
AsRh	mp-22079	8 5 4 (G)	52
Te2Au	mp-567525	7 7 5 (G)	17
Be3N2	mp-6977	10 10 3 (G)	46
Y2O3	mp-558573	8 4 3 (G)	96
AgO	mp-1079720	8 5 5 (G)	52
FeTe2	mp-1102002	4 4 4 (M)	68
H2O	mp-2739191	7 7 4 (G)	24
CdO2	mp-2310	5 5 5 (G)	68
Mo3Se4	mp-21021	4 4 4 (M)	86
PbSe	mp-22009	6 6 2 (G)	26
Be12Pd	mp-12666	6 6 6 (M)	69
Sr2As	mp-15697	6 6 3 (G)	28
AsRh2	mp-1102364	7 4 3 (G)	88
ZrRh	mp-2808	8 8 8 (M)	19
K2S	mp-559278	5 5 4 (G)	28
Ge4Rh	mp-1104286	4 4 3 (G)	135
Theoretical, metastable ($n = 60$)			
Li3Ag	mp-976408	7 7 7 (G)	24
Y3Co	mp-1189329	3 4 4 (G)	156
TiNi	mp-1067248	10 7 6 (G)	36
TaP	mp-1187244	9 9 9 (G)	13
Sr3As4	mp-678007	3 3 4 (G)	62
Ga2Se3	mp-1224809	5 5 5 (G)	30
MnN	mp-1009130	10 10 10 (G)	13
BeCl2	mp-1190080	3 3 3 (G)	78
MgSn	mp-1094801	8 8 6 (M)	13
TiW	mp-1216621	10 10 6 (M)	18
SrTi3	mp-1206636	5 5 5 (G)	32
IrCl3	mp-729507	3 3 6 (G)	84
Ta4C3	mp-1218000	6 6 6 (M)	48
Sc2O3	mp-755313	6 5 6 (G)	64
SnBr2	mp-978985	4 4 6 (M)	34
Sn7Pd5	mp-1219006	5 5 3 (G)	108
Tl4Sn	mp-1216560	5 5 5 (G)	45
LiTi3	mp-1185253	6 6 6 (M)	32
PtCl2	mp-1219748	5 5 4 (G)	34
HCl	mp-684609	7 7 7 (G)	5
MgSn	mp-1094669	8 8 8 (M)	13
K5As4	mp-684799	4 4 3 (G)	41
KSn3	mp-1185151	4 4 5 (G)	64
NbS2	mp-774873	10 6 2 (M)	34
YAg3	mp-1187811	6 6 6 (G)	37
B3Ir2	mp-1228647	9 5 4 (G)	60
TiMo2	mp-1216675	5 6 14 (G)	27
Cr2C	mp-1226378	10 10 7 (G)	22
ZrO2	mp-775909	10 6 4 (M)	36
Zn	mp-2647117	11 11 11 (G)	9
IN2	mp-1097011	5 5 5 (G)	24

Table S3 (continued)

Formula	mp-id	k-point grid	NBANDS
CoSe2	mp-1390196	4 4 4 (G)	68
GeS	mp-12910	7 7 5 (G)	26
ZrTi	mp-1215236	9 9 6 (G)	19
VCo3	mp-1095057	7 6 6 (G)	72
Tl3Ga	mp-1187539	5 5 5 (G)	36
CaGe2	mp-643542	7 7 6 (G)	23
NbRh2	mp-1220414	10 6 4 (M)	54
TcW	mp-1217337	10 10 6 (M)	18
MgF2	mp-560236	6 6 5 (G)	24
VN	mp-1017532	11 11 5 (G)	26
Na2Cl	mp-990084	6 6 6 (M)	12
ZnPb3	mp-1187949	5 5 5 (G)	36
Sr3Sn	mp-1187136	5 5 5 (G)	24
TaRe	mp-1217894	10 10 6 (M)	18
Al4Si	mp-1228120	6 6 6 (M)	20
B6Pb	mp-1096825	6 6 6 (M)	33
ScTc3	mp-867262	7 7 7 (G)	37
CrMo	mp-1226200	11 11 7 (G)	18
BPd5	mp-1228665	6 6 6 (M)	49
Zr3Be	mp-1188016	6 6 6 (M)	35
Na3Cl	mp-1064484	8 8 3 (G)	16
SiAu3	mp-1186998	7 7 7 (G)	31
InSe2	mp-1232036	7 7 3 (G)	34
TiSi2	mp-570991	8 8 8 (M)	17
BeAu3	mp-983539	7 7 7 (G)	32
Zr3Ge	mp-1188039	4 4 6 (G)	78
AlSb	mp-1023918	6 6 5 (G)	16
Mg15B	mp-1023515	4 4 3 (G)	64
HfMo	mp-1224314	10 10 6 (M)	18

E Elements and Allotropes

Every single-element structure with no matched polymorph in the dataset: the elemental references (**ELEMENT_REFERENCE**) used to decompose multi-element compounds. An experimental compound’s formula is marked with a superscript star (black if it is the stable phase, red if it is a metastable polymorph/allotrope); unmarked formulas are theoretical. Auto-generated by `report/gen_appendix_extension.py`.

Table S4: Elements and allotropes with no matched polymorph in the **extension** campaign: references used to decompose multi-element compounds. Allotropes with one or more other allotropes computed in this dataset (e.g. carbon) are grouped in Table S5 instead. Black * = stable experimental reference (thermodynamic phase); red * = metastable experimental reference; unmarked = theoretical. No Δ ICOHP column: an element is not itself a decomposition reaction.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
Ag*	mp-124	Fm-3m	0.0021
Al*	mp-134	Fm-3m	0.0000
As*	mp-158	Cmce	0.0000
Au*	mp-81	Fm-3m	0.0000

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
B*	mp-160	R-3m	0.0000
Ba*	mp-122	Im-3m	0.0000
Be*	mp-87	P6 ₃ /mmc	0.0000
Br*	mp-23154	Cmce	0.0000
Ca*	mp-21	Im-3m	0.0056
Cd*	mp-94	P6 ₃ /mmc	0.0000
Cl ₂ *	mp-22848	Cmce	0.0000
Co*	mp-102	Fm-3m	0.0000
Cr*	mp-90	Im-3m	0.0000
Cs*	mp-1	Im-3m	0.0193
Cu*	mp-30	Fm-3m	0.0000
F ₂ *	mp-561203	C2/c	0.0031
Fe*	mp-13	Im-3m	0.0000
Ga*	mp-142	Cmce	0.0000
Ge*	mp-32	Fd-3m	0.0000
H ₂ *	mp-730101	P2 ₁₂ ₁₂ ₁	0.0000
Hf*	mp-103	P6 ₃ /mmc	0.0000
Hg*	mp-10861	P6 ₃ /mmm	0.0030
I*	mp-23153	Cmce	0.0000
In*	mp-1055994	I4 ₁ /mmm	0.0045
Ir*	mp-101	Fm-3m	0.0000
Li*	mp-1018134	R-3m	0.0000
Mg*	mp-153	P6 ₃ /mmc	0.0000
Mn*	mp-35	I-43m	0.0000
Mo*	mp-129	Im-3m	0.0000
N ₂	mp-1059834	Imma	1.1010
Nb*	mp-75	Im-3m	0.0000
Ni*	mp-23	Fm-3m	0.0000
O ₂ *	mp-1524462	C2/m	0.0000
Os*	mp-49	P6 ₃ /mmc	0.0000
P*	mp-568348	P2 ₁ /c	0.0000
Pb*	mp-20483	Fm-3m	0.0000
Pd*	mp-2	Fm-3m	0.0000
Pt*	mp-126	Fm-3m	0.0000
Rb*	mp-70	Im-3m	0.0092
Re*	mp-8	P6 ₃ /mmc	0.0036
Rh*	mp-74	Fm-3m	0.0000
Ru*	mp-33	P6 ₃ /mmc	0.0000
S*	mp-77	Fddd	0.0000
Sb*	mp-104	R-3m	0.0000
Sc*	mp-67	P6 ₃ /mmc	0.0000
Se*	mp-14	P3 ₁ 21	0.0011
Sn*	mp-623511	I4 ₁ /mmm	0.0614
Sr*	mp-139	P6 ₃ /mmc	0.0000
Ta*	mp-50	Im-3m	0.0091
Tc*	mp-113	P6 ₃ /mmc	0.0000
Te*	mp-19	P3 ₁ 21	0.0000
Ti*	mp-46	P6 ₃ /mmc	0.0152
Tl*	mp-82	P6 ₃ /mmc	0.0000
V*	mp-146	Im-3m	0.0000
W*	mp-91	Im-3m	0.0000
Y*	mp-112	P6 ₃ /mmc	0.0000
Zn*	mp-79	P6 ₃ /mmc	0.0000
Zr*	mp-131	P6 ₃ /mmc	0.0000

F Polymorphs and Allotropes

Elements and compounds with several entries sharing a formula in the dataset, grouped here rather than scattered across the elements/compounds tables (see the table caption for detail).

Table S5: Elements (allotropes) and compounds (polymorphs) with **several entries sharing a formula** in the **extension** campaign (54 groups, 130 entries) — grouped here rather than scattered across the elements/compounds tables, since the natural comparison for a polymorph group is polymorph-to-polymorph, not the decomposition-to-elements reaction (hence no ΔICOHP column). A rule separates each group; within a group, sorted by increasing E_{hull} . Black * = stable experimental; red * = metastable experimental; unmarked = theoretical.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
AgN*	mp-1091417	Cmce	1.5201
AgN	mp-1009766	F-43m	1.5495
AgN	mp-1428312	Pm-3m	1.6519
AgN	mp-1424734	Ama2	1.9196
BaF ₃ *	mp-1080821	P2 ₁ /m	0.0000
BaF ₃	mp-1183303	I4/mmm	0.0612
BaO*	mp-1008500	P6 ₃ /mmc	0.0189
BaO	mp-1950989	P6 ₃ /mmc	0.0434
Be ₃ N ₂ *	mp-18337	Ia-3	0.0000
Be ₃ N ₂	mp-1017550	P-3m1	0.2081
Be ₃ P ₂ *	mp-567841	Ia-3	0.0000
Be ₃ P ₂	mp-1084771	Pn-3m	0.1567
BeCl ₂ *	mp-23267	Ibam	0.0087
BeCl ₂	mp-1172909	I-43m	3.1688
BeF ₂ *	mp-15951	P3 ₁ 21	0.0004
BeF ₂	mp-975644	P6/mmm	2.5181
BeO	mp-1778	F-43m	0.0070
BeO*	mp-7599	P4 ₂ /mnm	0.0153
Bi ₂ O ₃ *	mp-558953	P2 ₁ /c	0.6000
Bi ₂ O ₃	mp-674644	P1	0.6199
Bi ₂ O ₃	mp-1383505	P1	0.9068
Bi ₂ O ₃	mp-714859	P1	2.4029
C*	mp-48	P6 ₃ /mmc	0.0031
C*	mp-66	Fd-3m	0.1123
C*	mp-47	P6 ₃ /mmc	0.1395
C	mp-1190171	Pnma	0.2927
C	mp-1080826	C2/m	0.3009
C ₃ N ₄ *	mp-571653	P-43m	0.4981
C ₃ N ₄	mp-2852	I-43d	0.5033
C ₃ N ₄	mp-1182484	P6 ₃ /m	0.5848
C ₃ N ₄	mp-563	Fd-3m	1.5119
Ca ₃ N ₂ *	mp-844	Ia-3	0.0000
Ca ₃ N ₂ *	mp-1047	R-3c	0.0167
Ca ₃ N ₂	mp-1013524	Pm-3m	0.5112
CaCl ₂ *	mp-23214	Pnnm	0.0011

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
CaCl ₂	mp-1120739	Fm-3m	0.0422
CaF ₂ *	mp-10464	Pnma	0.0346
CaF ₂	mp-554355	P4/mmm	0.2403
CaO*	mp-2605	Fm-3m	0.0000
CaO*	mp-1079707	Cmce	0.2098
CaO	mp-1181903	Fd-3m	2.7281
CsN ₃ *	mp-510557	I4/mcm	0.0000
CsN ₃	mp-1225892	P4/mmm	0.0113
CsO ₂ *	mp-1441	I4/mmm	0.0349
CsO ₂	mp-1096936	Cmmm	0.0940
CsP ₇ *	mp-1201790	Pbcm	0.0000
CsP ₇	mp-1213206	Pca2_1	0.0001
K ₂ O*	mp-971	Fm-3m	0.0000
K ₂ O	mp-1223784	P4/mmm	0.2000
KF ₃	mp-2749823	P2_1	0.0119
KF ₃ *	mp-1544557	P2_12_12_1	0.0125
KN ₂ *	mp-1071868	Cmc2_1	1.1630
KN ₂	mp-1180757	Cmcm	1.2293
KP*	mp-1058315	R-3m	1.0080
KP	mp-1057787	Immm	1.0092
Li ₂ O*	mp-1960	Fm-3m	0.0000
Li ₂ O	mp-755894	Pnma	0.0844
Li ₂ S*	mp-1125	Pnma	0.0615
Li ₂ S	mp-557142	P6_3/mmc	0.1750
LiCl	mp-1185319	P6_3mc	0.0039
LiCl*	mp-22905	Fm-3m	0.0243
LiF*	mp-1138	Fm-3m	0.0000
LiF	mp-1009009	Pm-3m	0.2875
LiP ₅ *	mp-2412	Pna2_1	0.0077
LiP ₅	mp-32760	Pna2_1	0.0963
MgCl ₂ *	mp-23210	R-3m	0.0000
MgCl ₂	mp-569626	Ama2	0.1991
MgF ₂ *	mp-1072956	Pnnm	0.0025
MgF ₂	mp-1062842	Amm2	0.2636
MgO*	mp-549706	P6_3mc	0.1180
MgO	mp-1009127	Pm-3m	0.7484
MgS*	mp-1315	Fm-3m	0.0000
MgS	mp-13032	F-43m	0.0343
Na ₂ O*	mp-2352	Fm-3m	0.0000
Na ₂ O	mp-1975297	C2/m	0.0981
NaN ₃ *	mp-22003	R-3m	0.0017
NaN ₃	mp-1065265	C2/m	1.0162
NaS*	mp-409	P-62m	0.0023
NaS	mp-1180106	C2/m	1.0061
NbO*	mp-2692	Fm-3m	0.7888
NbO	mp-634965	P4/mmm	0.8240

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
NbP [*]	mp-9830	I4_1/amd	0.6021
NbP	mp-1220317	Cmmm	0.6209
OsC [*]	mp-7142	P-6m2	0.9161
OsC	mp-999308	P6_3/mmc	0.9278
OsC	mp-1009539	Fm-3m	1.3918
OsC	mp-1009537	Pm-3m	1.4795
Rb ₂ O [*]	mp-1394	Fm-3m	0.0032
Rb ₂ O	mp-2023482	P-3m1	0.0479
RbF	mp-11718	Fm-3m	0.0000
RbF [*]	mp-2064	Pm-3m	0.0516
RbN ₃ [*]	mp-581833	P4/mmm	0.0193
RbN ₃	mp-1219567	Amm2	0.7966
RbP [*]	mp-1179688	C2/m	0.9879
RbP	mp-1058499	Immm	0.9882
RbS [*]	mp-558071	Immm	0.0019
RbS	mp-1179684	C2/m	0.9975
ReC [*]	mp-1079494	P6_3/mmc	0.6362
ReC	mp-1009731	Pm-3m	1.1150
RuC [*]	mp-10030	P-6m2	0.6494
RuC	mp-1009787	Fm-3m	0.9097
SN [*]	COD:7017102	P2_1/c	—
SN [*]	mp-236	P2_1/c	0.5087
SN	mp-684642	P2_1	0.5498
SN	mp-1173453	P2_1	0.7225
SN	mp-1078816	P2_1/c	0.8090
SiO ₂ [*]	mp-9258	Pa-3	0.5770
SiO ₂	mp-10064	Fm-3m	1.2779
SiO ₂	mp-556044	P-31c	1.4574
SiO ₂	mp-638049	F4_132	1.7448
Sr ₃ P ₄ [*]	mp-14288	Fdd2	0.0000
Sr ₃ P ₄	mp-682953	P1	0.0081
SrF ₂	mp-1102911	Pnma	0.0267
SrF ₂ [*]	mp-1102240	Pnma	0.0646
SrO ₂ [*]	mp-726705	Pnma	0.0096
SrO ₂	mp-2763179	Pnma	0.0110
TaN [*]	mp-1497	P6/mmm	0.4516
TaN	mp-1009833	F-43m	0.4917
TaN	mp-1009831	Pm-3m	0.8163
TiO ₂ [*]	mp-1439	Pbcn	0.0045
TiO ₂ [*]	mp-430	P2_1/c	0.0395
TiO ₂ [*]	mp-9173	Pnma	0.0575
TiC [*]	mp-567465	Fm-3m	2.1080
TiC	mp-1058887	P4/mmm	2.3757
TiN [*]	mp-1017982	P6_3mc	0.7334
TiN	mp-1007778	F-43m	0.7337
TiN	mp-1002222	Fm-3m	0.9712
WC [*]	mp-13136	Fm-3m	0.4477
WC	mp-1008635	F-43m	0.6729

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
WC	mp-1008630	Pm-3m	0.8990

G Target Compounds

Every multi-element target compound with no matched polymorph in the dataset, with ΔICOHP per atom and its endobondic/exobondic label where a case-1 decomposition reaction could be computed. Auto-generated from `analysis/reaction_icohp_case1.csv` and `analysis/reaction_analysis_ca` by `report/gen_appendix_extension.py`.

Table S6: Target compounds with no matched polymorph in the **extension** campaign. Formulas with several entries in this dataset are grouped in Table S5 instead. Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ΔICOHP per atom, products – reactants convention (`reaction_analysis`); “–” where no case-1 reaction could be computed (missing elemental reference).

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	$\Delta\text{ICOHP}/\text{at.}$ (eV)	Label
Ba ₅ P ₄ *	mp-17566	Pnma	0.0000	-0.9590	exobondic
BaCl ₂ *	mp-23199	Pnma	0.0068	-0.2244	exobondic
BaN ₂ *	mp-1102371	Cc	2.0568	-8.6248	exobondic
BaS ₃ *	mp-556296	P2_12_12	0.0436	-0.6233	exobondic
CaN*	mp-1058549	Fm-3m	0.3982	-4.9183	exobondic
Cs ₂ S*	mp-1077814	P6_3/mmc	0.0667	0.9554	endobondic
CsCl*	mp-573697	Fm-3m	0.0071	0.8830	endobondic
CsF*	mp-8455	Pm-3m	0.1088	3.8464	endobondic
K ₂ S*	mp-1022	Fm-3m	0.0000	-0.3359	exobondic
KCl*	mp-23289	Pm-3m	0.0776	0.7616	endobondic
KN ₃	mp-636056	I4/mcm	1.0370	-5.9991	exobondic
Li ₃ N*	mp-2341	P6_3/mmc	0.0039	0.6165	endobondic
Mn ₂ O ₇ *	mp-28338	P2_1/c	0.3346	-1.4687	exobondic
MnO ₂ *	mp-510408	P4_2/mnm	0.0464	-0.5274	exobondic
Na ₂ Cl	mp-1077015	Cmmm	0.2391	-0.3965	exobondic
PbN ₆ *	mp-667338	Pnma	0.1093	-2.5257	exobondic
RbCl*	mp-23299	Pm-3m	0.0298	1.7924	endobondic
S ₂ N*	COD:4031496	P4_2nm	–	-0.9767	exobondic
SrN*	mp-1058586	Fm-3m	0.4343	-4.5395	exobondic
SrS*	mp-10627	Pm-3m	0.2996	-0.1088	exobondic

H Endobondic Reactions

The subset of case-1 decomposition reactions ($A_a B_b \rightarrow a A + b B$, standard-state elemental references, balanced by `reaction_icohp.py`) labeled **endobondic** ($\Delta\text{ICOHP} \geq 0$, products – reactants convention — note this is the sign *opposite* `reaction_icohp.py`’s own `delta_icohp_per_atom` column reported elsewhere in this project; see `delta.py`’s docstring).

Table S7: Decomposition-into-elements (case 1) reactions labeled **endobondic** ($\Delta\text{ICOHP} \geq 0$, products – reactants) among the 208 compounds of the **extension** campaign. Black * = stable experimental starting compound; red * = metastable experimental.

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP}/\text{at.}$ (eV)
0.3333 Be ₃ N ₂ → Be + 0.3333 N ₂	mp-1017550	0.2081	0.3960
BeCl ₂ extcoloured* → Be + Cl ₂	mp-23267	0.0087	1.8204
BeF ₂ extcoloured* → Be + F ₂	mp-15951	0.0004	3.1514
BeO extcoloured* → Be + 0.5 O ₂	mp-7599	0.0153	2.7864
BeO → Be + 0.5 O ₂	mp-1778	0.0070	4.4391
0.3333 C ₃ N ₄ extcoloured* → C + 0.6667 N ₂	mp-571653	0.4981	1.1208
0.3333 C ₃ N ₄ → C + 0.6667 N ₂	mp-2852	0.5033	1.8456
0.5 Cs ₂ S extcoloured* → Cs + 0.5 S	mp-1077814	0.0667	0.9554
CsCl extcoloured* → Cs + 0.5 Cl ₂	mp-573697	0.0071	0.8830
CsF extcoloured* → Cs + 0.5 F ₂	mp-8455	0.1088	3.8464
CsO ₂ extcoloured* → Cs + O ₂	mp-1441	0.0349	0.0666
CsO ₂ → Cs + O ₂	mp-1096936	0.0940	0.7743
KCl extcoloured* → K + 0.5 Cl ₂	mp-23289	0.0776	0.7616
0.5 Li ₂ O* → Li + 0.25 O ₂	mp-1960	0.0000	2.3366
0.5 Li ₂ O → Li + 0.25 O ₂	mp-755894	0.0844	1.9481
0.5 Li ₂ S extcoloured* → Li + 0.5 S	mp-1125	0.0615	1.8505
0.5 Li ₂ S → Li + 0.5 S	mp-557142	0.1750	1.5052
0.3333 Li ₃ N extcoloured* → Li + 0.1667 N ₂	mp-2341	0.0039	0.6165
LiCl extcoloured* → Li + 0.5 Cl ₂	mp-22905	0.0243	2.8030
LiCl → Li + 0.5 Cl ₂	mp-1185319	0.0039	2.5005
LiF* → Li + 0.5 F ₂	mp-1138	0.0000	4.4673
LiF → Li + 0.5 F ₂	mp-1009009	0.2875	4.5131
LiP ₅ extcoloured* → Li + 5 P	mp-2412	0.0077	0.9043
LiP ₅ → Li + 5 P	mp-32760	0.0963	0.8692
MgCl ₂ * → Mg + Cl ₂	mp-23210	0.0000	0.1568
MgF ₂ extcoloured* → Mg + F ₂	mp-1072956	0.0025	0.0015
MgF ₂ → Mg + F ₂	mp-1062842	0.2636	0.0229
0.5 Rb ₂ O extcoloured* → Rb + 0.25 O ₂	mp-1394	0.0032	0.7235
RbCl extcoloured* → Rb + 0.5 Cl ₂	mp-23299	0.0298	1.7924
RbF extcoloured* → Rb + 0.5 F ₂	mp-2064	0.0516	3.9244
RbF → Rb + 0.5 F ₂	mp-11718	0.0000	1.9497
SiO ₂ extcoloured* → Si + O ₂	mp-9258	0.5770	3.3332
SiO ₂ → Si + O ₂	mp-10064	1.2779	2.1735
SiO ₂ → Si + O ₂	mp-556044	1.4574	0.0958
SiO ₂ → Si + O ₂	mp-638049	1.7448	0.3849

I Exobondic Reactions

The complementary **exobondic** subset ($\Delta\text{ICOHP} < 0$), same convention and provenance as Appendix H.

Table S8: Decomposition-into-elements (case 1) reactions labeled **exobondic** ($\Delta\text{ICOHP} < 0$, products – reactants) among the 208 compounds of the **extension** campaign. Black * = stable experimental starting compound; red * = metastable experimental.

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP}/\text{at.}$ (eV)
AgN extcoloured* → Ag + 0.5 N ₂	mp-1091417	1.5201	-7.5875
AgN → Ag + 0.5 N ₂	mp-1009766	1.5495	-6.4957

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	Δ ICOHP/at. (eV)
$\text{AgN} \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1424734	1.9196	-8.0907
$\text{AgN} \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1428312	1.6519	-5.3559
$0.2 \text{ Ba}_5\text{P}_4 \text{ extcoloured}^* \rightarrow \text{Ba} + 0.8 \text{ P}$	mp-17566	0.0000	-0.9590
$\text{BaCl}_2 \text{ extcoloured}^* \rightarrow \text{Ba} + \text{Cl}_2$	mp-23199	0.0068	-0.2244
$\text{BaF}_3^* \rightarrow \text{Ba} + 1.5 \text{ F}_2$	mp-1080821	0.0000	-0.8680
$\text{BaF}_3 \rightarrow \text{Ba} + 1.5 \text{ F}_2$	mp-1183303	0.0612	-0.2493
$\text{BaN}_2 \text{ extcoloured}^* \rightarrow \text{Ba} + \text{N}_2$	mp-1102371	2.0568	-8.6248
$\text{BaO} \text{ extcoloured}^* \rightarrow \text{Ba} + 0.5 \text{ O}_2$	mp-1008500	0.0189	-0.3221
$\text{BaO} \rightarrow \text{Ba} + 0.5 \text{ O}_2$	mp-1950989	0.0434	-1.6129
$\text{BaS}_3 \text{ extcoloured}^* \rightarrow \text{Ba} + 3 \text{ S}$	mp-556296	0.0436	-0.6233
$0.3333 \text{ Be}_3\text{N}_2^* \rightarrow \text{Be} + 0.3333 \text{ N}_2$	mp-18337	0.0000	-0.0746
$0.3333 \text{ Be}_3\text{P}_2^* \rightarrow \text{Be} + 0.6667 \text{ P}$	mp-567841	0.0000	-0.2111
$0.3333 \text{ Be}_3\text{P}_2 \rightarrow \text{Be} + 0.6667 \text{ P}$	mp-1084771	0.1567	-0.1795
$\text{BeCl}_2 \rightarrow \text{Be} + \text{Cl}_2$	mp-1172909	3.1688	-5.7807
$\text{BeF}_2 \rightarrow \text{Be} + \text{F}_2$	mp-975644	2.5181	-0.6816
$0.3333 \text{ C}_3\text{N}_4 \rightarrow \text{C} + 0.6667 \text{ N}_2$	mp-1182484	0.5848	-3.6399
$0.3333 \text{ C}_3\text{N}_4 \rightarrow \text{C} + 0.6667 \text{ N}_2$	mp-563	1.5119	-0.4498
$0.3333 \text{ Ca}_3\text{N}_2 \text{ extcoloured}^* \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-1047	0.0167	-4.4881
$0.3333 \text{ Ca}_3\text{N}_2^* \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-844	0.0000	-4.4374
$0.3333 \text{ Ca}_3\text{N}_2 \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-1013524	0.5112	-4.4898
$\text{CaCl}_2 \text{ extcoloured}^* \rightarrow \text{Ca} + \text{Cl}_2$	mp-23214	0.0011	-0.3199
$\text{CaCl}_2 \rightarrow \text{Ca} + \text{Cl}_2$	mp-1120739	0.0422	-0.2208
$\text{CaF}_2 \text{ extcoloured}^* \rightarrow \text{Ca} + \text{F}_2$	mp-10464	0.0346	-0.2227
$\text{CaF}_2 \rightarrow \text{Ca} + \text{F}_2$	mp-554355	0.2403	-0.1814
$\text{CaN} \text{ extcoloured}^* \rightarrow \text{Ca} + 0.5 \text{ N}_2$	mp-1058549	0.3982	-4.9183
$\text{CaO} \text{ extcoloured}^* \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-1079707	0.2098	-2.2127
$\text{CaO}^* \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-2605	0.0000	-0.9525
$\text{CaO} \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-1181903	2.7281	-2.9971
$\text{CsN}_3^* \rightarrow \text{Cs} + 1.5 \text{ N}_2$	mp-510557	0.0000	-2.2903
$\text{CsN}_3 \rightarrow \text{Cs} + 1.5 \text{ N}_2$	mp-1225892	0.0113	-1.6664
$\text{CsP}_7^* \rightarrow \text{Cs} + 7 \text{ P}$	mp-1201790	0.0000	-0.2978
$\text{CsP}_7 \rightarrow \text{Cs} + 7 \text{ P}$	mp-1213206	0.0001	-0.2667
$0.5 \text{ K}_2\text{O}^* \rightarrow \text{K} + 0.25 \text{ O}_2$	mp-971	0.0000	-0.4357
$0.5 \text{ K}_2\text{O} \rightarrow \text{K} + 0.25 \text{ O}_2$	mp-1223784	0.2000	-1.5638
$0.5 \text{ K}_2\text{S}^* \rightarrow \text{K} + 0.5 \text{ S}$	mp-1022	0.0000	-0.3359
$\text{KF}_3 \text{ extcoloured}^* \rightarrow \text{K} + 1.5 \text{ F}_2$	mp-1544557	0.0125	-1.1511
$\text{KF}_3 \rightarrow \text{K} + 1.5 \text{ F}_2$	mp-2749823	0.0119	-1.1505
$\text{KN}_2 \text{ extcoloured}^* \rightarrow \text{K} + \text{N}_2$	mp-1071868	1.1630	-3.4244
$\text{KN}_2 \rightarrow \text{K} + \text{N}_2$	mp-1180757	1.2293	-4.2865
$\text{KN}_3 \rightarrow \text{K} + 1.5 \text{ N}_2$	mp-636056	1.0370	-5.9991
$\text{KP} \text{ extcoloured}^* \rightarrow \text{K} + \text{P}$	mp-1058315	1.0080	-2.1600
$\text{KP} \rightarrow \text{K} + \text{P}$	mp-1057787	1.0092	-2.1366
$\text{MgCl}_2 \rightarrow \text{Mg} + \text{Cl}_2$	mp-569626	0.1991	-0.0862
$\text{MgO} \text{ extcoloured}^* \rightarrow \text{Mg} + 0.5 \text{ O}_2$	mp-549706	0.1180	-1.4179
$\text{MgO} \rightarrow \text{Mg} + 0.5 \text{ O}_2$	mp-1009127	0.7484	-1.0763
$\text{MgS}^* \rightarrow \text{Mg} + \text{S}$	mp-1315	0.0000	-0.3401
$\text{MgS} \rightarrow \text{Mg} + \text{S}$	mp-13032	0.0343	-0.4163
$0.5 \text{ Mn}_2\text{O}_7 \text{ extcoloured}^* \rightarrow \text{Mn} + 1.75 \text{ O}_2$	mp-28338	0.3346	-1.4687
$\text{MnO}_2 \text{ extcoloured}^* \rightarrow \text{Mn} + \text{O}_2$	mp-510408	0.0464	-0.5274
$0.5 \text{ Na}_2\text{Cl} \rightarrow \text{Na} + 0.25 \text{ Cl}_2$	mp-1077015	0.2391	-0.3965
$0.5 \text{ Na}_2\text{O}^* \rightarrow \text{Na} + 0.25 \text{ O}_2$	mp-2352	0.0000	-1.3057
$0.5 \text{ Na}_2\text{O} \rightarrow \text{Na} + 0.25 \text{ O}_2$	mp-1975297	0.0981	-1.5619
$\text{NaN}_3 \text{ extcoloured}^* \rightarrow \text{Na} + 1.5 \text{ N}_2$	mp-22003	0.0017	-1.5201
$\text{NaN}_3 \rightarrow \text{Na} + 1.5 \text{ N}_2$	mp-1065265	1.0162	-5.4248
$\text{NaS} \text{ extcoloured}^* \rightarrow \text{Na} + \text{S}$	mp-409	0.0023	-0.4911
$\text{NaS} \rightarrow \text{Na} + \text{S}$	mp-1180106	1.0061	-2.3999

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)
NbO extcolored* $\rightarrow$ Nb + 0.5 O ₂	mp-2692	0.7888	-2.3179
NbO $\rightarrow$ Nb + 0.5 O ₂	mp-634965	0.8240	-3.1495
NbP extcolored* $\rightarrow$ Nb + P	mp-9830	0.6021	-2.4724
NbP $\rightarrow$ Nb + P	mp-1220317	0.6209	-1.1731
OsC extcolored* $\rightarrow$ Os + C	mp-7142	0.9161	-2.0168
OsC $\rightarrow$ Os + C	mp-1009537	1.4795	-2.3184
OsC $\rightarrow$ Os + C	mp-1009539	1.3918	-3.1141
OsC $\rightarrow$ Os + C	mp-999308	0.9278	-2.9572
PbN ₆ extcolored* $\rightarrow$ Pb + 3 N ₂	mp-667338	0.1093	-2.5257
0.5 Rb ₂ O $\rightarrow$ Rb + 0.25 O ₂	mp-2023482	0.0479	-0.7592
RbN ₃ extcolored* $\rightarrow$ Rb + 1.5 N ₂	mp-581833	0.0193	-1.3478
RbN ₃ $\rightarrow$ Rb + 1.5 N ₂	mp-1219567	0.7966	-3.5636
RbP extcolored* $\rightarrow$ Rb + P	mp-1179688	0.9879	-1.9315
RbP $\rightarrow$ Rb + P	mp-1058499	0.9882	-1.9104
RbS extcolored* $\rightarrow$ Rb + S	mp-558071	0.0019	-0.3217
RbS $\rightarrow$ Rb + S	mp-1179684	0.9975	-2.7347
ReC extcolored* $\rightarrow$ Re + C	mp-1079494	0.6362	-2.6625
ReC $\rightarrow$ Re + C	mp-1009731	1.1150	-1.9955
RuC extcolored* $\rightarrow$ Ru + C	mp-10030	0.6494	-2.1301
RuC $\rightarrow$ Ru + C	mp-1009787	0.9097	-2.8605
0.5 S ₂ N extcolored* $\rightarrow$ S + 0.25 N ₂	COD:4031496	—	-0.9767
SN extcolored* $\rightarrow$ S + 0.5 N ₂	COD:7017102	—	-1.3963
SN extcolored* $\rightarrow$ S + 0.5 N ₂	mp-236	0.5087	-2.3179
SN $\rightarrow$ S + 0.5 N ₂	mp-1078816	0.8090	-3.7082
SN $\rightarrow$ S + 0.5 N ₂	mp-1173453	0.7225	-3.1673
SN $\rightarrow$ S + 0.5 N ₂	mp-684642	0.5498	-1.7362
0.3333 Sr ₃ P ₄ * $\rightarrow$ Sr + 1.333 P	mp-14288	0.0000	-1.0472
0.3333 Sr ₃ P ₄ $\rightarrow$ Sr + 1.333 P	mp-682953	0.0081	-1.0793
SrF ₂ extcolored* $\rightarrow$ Sr + F ₂	mp-1102240	0.0646	-0.5421
SrF ₂ $\rightarrow$ Sr + F ₂	mp-1102911	0.0267	-0.1481
SrN extcolored* $\rightarrow$ Sr + 0.5 N ₂	mp-1058586	0.4343	-4.5395
SrO ₂ extcolored* $\rightarrow$ Sr + O ₂	mp-726705	0.0096	-1.9205
SrO ₂ $\rightarrow$ Sr + O ₂	mp-2763179	0.0110	-1.7522
SrS extcolored* $\rightarrow$ Sr + S	mp-10627	0.2996	-0.1088
TaN extcolored* $\rightarrow$ Ta + 0.5 N ₂	mp-1497	0.4516	-5.9309
TaN $\rightarrow$ Ta + 0.5 N ₂	mp-1009831	0.8163	-4.0540
TaN $\rightarrow$ Ta + 0.5 N ₂	mp-1009833	0.4917	-5.3385
TiO ₂ extcolored* $\rightarrow$ Ti + O ₂	mp-1439	0.0045	-0.6461
TiO ₂ extcolored* $\rightarrow$ Ti + O ₂	mp-9173	0.0575	-0.6509
TiO ₂ extcolored* $\rightarrow$ Ti + O ₂	mp-430	0.0395	-0.6267
TlC extcolored* $\rightarrow$ Tl + C	mp-567465	2.1080	-3.1404
TlC $\rightarrow$ Tl + C	mp-1058887	2.3757	-3.3638
TlN extcolored* $\rightarrow$ Tl + 0.5 N ₂	mp-1017982	0.7334	-3.7498
TlN $\rightarrow$ Tl + 0.5 N ₂	mp-1002222	0.9712	-2.3228
TlN $\rightarrow$ Tl + 0.5 N ₂	mp-1007778	0.7337	-3.1948
WC extcolored* $\rightarrow$ W + C	mp-13136	0.4477	-6.2684
WC $\rightarrow$ W + C	mp-1008630	0.8990	-5.9400
WC $\rightarrow$ W + C	mp-1008635	0.6729	-7.7315

J ICOHP and ICOBI Antibonding Descriptors

J.1 Implementation

The ICOHP antibonding-population-near-frontier descriptor (§6.2, `cohpy_extraction.py`) is extended here to ICOBI: same one-sided window ($E_{\text{ref}} - \Delta E$, E_{ref}], same E_{ref} (Fermi level for metals,

VBM for gapped compounds — logic entirely independent of which curve is integrated), read this time from `COBICAR.lobster/ICOBILIST.lobster` instead of `COHPCAR.lobster/ICOHPLIST.lobster` (`pymatgen.io.lobster.outputs.Cohpcar(are_cobis=True)`).

Critical point, empirically verified, not assumed: ICOBI does **not** share ICOHP's sign convention. Whereas "negative ICOHP = bonding" (§2.4), ICOBI follows the opposite convention, "positive ICOBI = bonding" (like COOP, not like COHP). Verified two ways on real data: (1) exact cross-check against `ICOBILIST.lobster` (max deviation 0.0 across 66 bond labels, `tests/test_cohp_extraction.py`) — `pymatgen` does not silently renormalize the sign when switching from COHP to COBI parsing; (2) on `extension_SiO2_anchor`'s strongest Si–O bond (ICOHP = -5.599 , ICOBI = $+0.513$), the deep bonding region (-10 to -3 eV, far from any frontier effects) gives a mean COHP(E) of -0.338 but a mean COBI(E) of $+0.041$ — opposite signs in the same physically bonding-dominated region. ICOBI's antibonding part is therefore its *negative* part, not its positive part – the clip direction (`integrate_icobi_antibonding_in_window()`) is deliberately the mirror image of the ICOHP version, not a copy.

Computed across the 394 compounds with a `COBICAR.lobster` file (`analysis/compute_icobi_antibonding.392` with `is_metal` available).

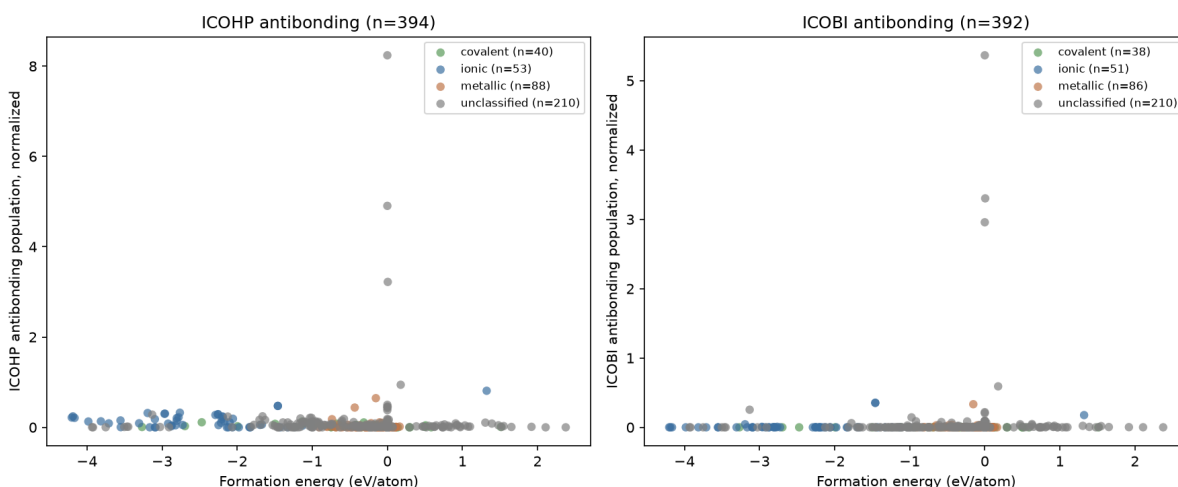
J.2 Correlation with formation energy

Table S9: Spearman correlations (normalized, $\Delta E = 1.0$ eV) against `formation_energy_per_atom` — ICOHP (n=394) and ICOBI (n=392) antibonding, same population.

Group	n (ICOHP)	ρ (ICOHP)	n (ICOBI)	ρ (ICOBI)
all	391	-0.2473^{***}	385	$+0.2518^{***}$
covalent	40	-0.4781^{**}	38	-0.1755
ionic	53	-0.1842	51	$+0.1289$
metallic	88	-0.1864	86	$+0.0011$
<code>is_metal=False</code>	166	-0.2621^{***}	162	$+0.0661$
<code>is_metal=True</code>	225	-0.0154	223	$+0.1263$

** $p < 0.01$; *** $p < 0.001$; no star = not significant. Full detail (raw + normalized, all three ΔE):

`analysis/stats_summary_antibonding_joint.json`.



The two descriptors behave differently. ICOBI antibonding's global sign is the *opposite* of ICOHP antibonding's ($+0.25$ vs. -0.25) — for ICOHP, more occupied antibonding character

near E_F is associated with *more* stable formation (already noted as counterintuitive in §6.2); for ICOBI, the relationship runs the naively expected way (more antibonding $\Rightarrow$ less stable). But **no ICOBI subgroup survives stratification** (covalent, ionic, metallic, `is_metal`: $p > 0.05$ throughout) — unlike ICOHP, whose covalent subgroup ($n = 40$, $\rho = -0.478$) remains the project’s most robust stratified result. ICOBI’s global correlation is therefore, like reaction ICOHP’s (§6.3), likely a between-bond-type clustering effect rather than a robust relationship within a homogeneous group.

J.3 $\Delta(\text{antibonding})$ of the decomposition-into-elements reaction

For a case-1 reaction (compound $\rightarrow$ elements), $\Delta(\text{antibonding})$ is **not** computed by `reaction_analysis/delta.py` (which combines *extensive* per-formula-unit ICOHP/ICOBI sums with integer coefficients): the antibonding population near the frontier is built from LOBSTER’s own "average" trace, an *intensive* quantity (averaged over all bonds, not summed) with no natural extensive decomposition. Convention chosen here, documented as an explicit methodological choice rather than a silent reuse: $\Delta = (\text{atomic-fraction-weighted average of the elements’ own antibonding descriptors}) - (\text{the compound’s own antibonding descriptor})$, still in products – reactants convention. Computed for 320 case-1 reactions (`analysis/compute_delta_antibonding_case1.py`).

Table S10: Spearman correlations, $\Delta(\text{antibonding})$ of the decomposition reaction (case 1, $n = 316\text{--}320$).

Descriptor	Target	n	ρ
$\Delta(\text{ICOHP antibonding})$	<code>formation_energy_per_atom</code>	316	-0.6439^{***}
$\Delta(\text{ICOBI antibonding})$	<code>formation_energy_per_atom</code>	316	-0.4457^{***}
$\Delta(\text{ICOHP antibonding})$	<code>energy_above_hull</code>	320	-0.1868^{***}
$\Delta(\text{ICOBI antibonding})$	<code>energy_above_hull</code>	320	-0.1524^{**}

$^{**}p < 0.01$; $^{***}p < 0.001$.

$\Delta(\text{ICOHP antibonding})$ is, at this stage, the strongest continuous correlation in the whole project ($\rho = -0.644$, stronger than reaction ICOHP in §6.3, $\rho = 0.502$) — to be read with caution: this is a newly chosen construction, not yet stratified by bond type for lack of time in this session, and its magnitude will need to reproduce before being treated as an established result. Notable: unlike the raw values (opposite signs, above), **both Δ s run the same direction** (both negative) — the delta construction appears to align the two descriptors on a consistent reading, where their raw values do not. The two Δ s are themselves correlated with each other ($\rho = 0.532$, $p < 0.001$, $n = 320$): related but not redundant.